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# Hourly Hypotension Early Warning Under Sparse MAP Charting: A Threshold Calibration Approach

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# Abstract

Sustained hypotension in Intensive Care Unit (ICU) settings, where a patient's blood pressure remains dangerously low over a prolonged period, is associated with organ damage and increased mortality rates. Early warning systems that flag deterioration give clinicians time to intervene. However building such a system on routinely charted Electronic Health Record (EHR) data raises a problem that standard evaluation practices overlook, that Mean Arterial Pressure (MAP), the single strongest predictor of a MAP-defined event, is absent in roughly 56% of patient hours. A model trained on blood pressure features will produce systematically lower risk scores during those unmonitored hours, and a globally calibrated alarm threshold can silently fail for more than half of all predictions.

We developed an hourly early warning system for sustained hypotension using non-waveform data from MIMIC-IV v3.1, covering 94,259 ICU stays on a fixed hourly grid over the first 48 hours of each admission. Sustained hypotension was defined using a window-based proxy (at least two hours with MAP below 65 mmHg within a rolling three hour window), adapted from minute-level benchmarks to tolerate sparse charting. A total of 353 features drawn from 20 clinical signal blocks were engineered, and two ablation suites totalling 30 model variants were evaluated under a fixed 10 percent alarm budget with thresholds frozen on the validation set before test evaluation.

Blood pressure features alone achieved an AUROC (a measure of how well the model separates high-risk from low-risk hours, where 1.0 is perfect) of 0.955 and caught 99.3% of events using 37 features. Non-BP features plateaued at an information ceiling around 0.75 AUROC regardless of how many signal families were added. Under the global threshold, MAP was absent in 55.7% of test hours, and the blood pressure model produced zero alarms during all of those hours, and all 32 MAP-missing events received no warning. Calibrating a separate threshold on the MAP-missing subset of the same model, without any retraining, recovered 93.75% recall on those 32 events (95% CI: 79.2%–99.2%), a figure confirmed as stable under bootstrap resampling of the threshold calibration (1,000 resamples). These 32 events are constrained by the label definition itself, which requires nearby MAP observations to confirm hypotension; truly unmonitored patients may not appear in the dataset at all. Our recommended configuration uses one lightweight tree model with two thresholds conditioned on MAP availability, achieving 97.6% overall event recall at 0.89 alerts per 24 hours. It outperformed all two-model gated alternatives while requiring no specialised hardware, making it operationally simple enough for the clinical settings where MAP absence is most common, pending prospective validation.

# Lay Abstract

When a patient's blood pressure drops and stays low in an intensive care unit (ICU), it can cause serious harm to kidneys, brain and other vital organs. If doctors and nurses could be warned a few hours before this occurs, they could act earlier to stabilise the patient. This project proposes a system that reads the information already recorded in a patient's medical chart (such as vital signs, lab results and medications) and checks every hour whether a dramatic drop in blood pressure is likely in the next three hours. If the risk is high enough, it ends up raising an alert.

One major challenge is that blood pressure itself is not recorded every hour for every patient. In the large hospital dataset used (covering over 94,000 ICU admissions), blood pressure was missing for more than half of all hourly records. When the system was tested against a single standard alarm setting, it worked well for the hours where blood pressure had been recorded recently but produced no alerts at all for the hours where it had not observed blood pressure. Patients who happened to be monitored less frequently were invisible to the alarm, not necessarily because they were safe, but because the alarm sensitivity was set for another group.

The proposed fix is that instead of setting one alarm for everyone, we used two. One standard setting for hours with recent blood pressure readings and a more sensitive setting for hours without. With this change in inference, the system caught over 97% of dangerous episodes while averaging less than one alert per day per patient. The model is a lightweight program with a lookup rule at inference, requiring no specialised hardware.

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# Contents

# List of Figures

# List of Tables

# Nomenclature

| | |
|---|---|
| AHE | Acute Hypotensive Episode |
| AUPRC | Area Under the Precision-Recall Curve |
| AUROC | Area Under the Receiver Operating Characteristic Curve |
| BP | Blood Pressure |
| BUN | Blood Urea Nitrogen |
| CBC | Complete Blood Count |
| CHF | Congestive Heart Failure |
| CI | Confidence Interval (Clopper–Pearson exact method) |
| CKD | Chronic Kidney Disease |
| COPD | Chronic Obstructive Pulmonary Disease |
| Config G | Deployment configuration using one model with two thresholds conditioned on MAP availability |
| DBP | Diastolic Blood Pressure |
| EHR | Electronic Health Record |
| FiO2 | Fraction of Inspired Oxygen |
| GCS | Glasgow Coma Scale |
| HGB | Histogram Gradient Boosting (scikit-learn HistGradientBoostingClassifier) |
| HR | Heart Rate |
| ICU | Intensive Care Unit |
| MAP | Mean Arterial Pressure |
| MIMIC-IV | Medical Information Mart for Intensive Care |
| NEE | Norepinephrine-Equivalent dose |
| NLR | Neutrophil-to-Lymphocyte Ratio |
| PEEP | Positive End-Expiratory Pressure |
| RR | Respiratory Rate |
| RRT | Renal Replacement Therapy |
| SBP | Systolic Blood Pressure |
| SF Ratio | SpO2-to-FiO2 Ratio |
| SpO2 | Peripheral Oxygen Saturation |
| TRIPOD | Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis |
| XGBoost | Extreme Gradient Boosting |

**Units**

| | |
|---|---|
| bpm | beats per minute |
| mmHg | millimetres of mercury |
| mL | millilitres |

# 1 Introduction

## 1.1 Problem Motivation

When a patient's blood pressure drops and stays low for an extended period in the Intensive Care Unit, the consequences can be severe. This condition, sustained hypotension, reduces blood flow to vital organs, and if it persists, can lead to kidney damage, neurological damage and increased risk of death [1, 2]. Clinicians use a measure called Mean Arterial Pressure (MAP), which represents the average pressure in the arteries in a single cardiac cycle, as one of the primary indicators of a patient's hemodynamic state. International guidelines for managing septic shock recommend maintaining MAP above 65 mmHg, reflecting how central this signal is to critical care decision making [3].

If clinicians could be warned several hours before sustained hypotension develops, they could intervene earlier to re-assess the patient before the situation worsens. An early warning system that produces a risk score every hour using data already recorded in a patient's health record, could provide this warning without requiring any additional monitoring equipment.

However building such a system is not the entire problem. In a real ICU setting, the clinical staff already face a high volume of alarms, many of which are non-actionable alarms. Alarm fatigue, which is when frequent low-value alerts lead to desensitisation and delayed responses, has been formally recognised as a patient safety concern [4, 5]. Any predictive model that generates too many alerts faces risk of contributing to alarm fatigue. So an early warning system must be evaluated on whether it can do so within an acceptable alarm budget or not.

There is a further complication that motivates this thesis. In non-waveform EHR data (data recorded through routine clinical charting rather than continuous monitor feeds), MAP is not always available. In the MIMIC-IV dataset used in this study, MAP is absent in approximately 56% of patient-hours. A model trained primarily on blood pressure features will produce different risk scores depending on if MAP was recently measured. If the alarm threshold is calibrated on the full population without accounting for this, the result can be that patients without recent MAP readings receive no alarms at all, not because they are low risk but because their monitoring pattern produces lower scores. This silent failure is the central problem that this thesis investigates.

## 1.2 Thesis Problem Statement

This thesis addresses the problem of hourly early warning for sustained hypotension in adult ICU patients, using retrospective non-waveform EHR data from MIMIC-IV v3.1, a large, freely available critical care dataset covering over 94,000 ICU stays at Beth Israel Deaconess Medical Center [6].

The prediction task is framed as follows, for each hour during the first 48 hours of an ICU stay, the model outputs a risk score estimating whether sustained hypotension will begin in the next three hours. Sustained hypotension is defined using a window-based persistence proxy, at least two hours with MAP below 65 mmHg within a rolling three hour window. This definition was adapted from minute-level benchmarks to tolerate the sparse charting patterns of hourly non-waveform data, where MAP may not be recorded for several consecutive hours.

The core challenge is that MAP, the signal on which the event definition is based and which turns out to be dominant predictor, is absent in more than half of all patient-hours. This introduces two problems. First is that the event definition must be constructed carefully to avoid depending on consecutive observations that the data cannot reliably provide. Second, any model that relies heavily on MAP features will produce systematically different scores for MAP-observed versus MAP-missing hours, and a globally calibrated alarm threshold may fail silently for the MAP-missing population.

This is a retrospective dataset study, not a clinical deployment. All experiments use historical data with patient-level train, validation, and test splits.

## 1.3 Research Aim and Objectives

The aim of this thesis is to build and evaluate an alarm-style early warning system for sustained hypotension that performs reliably when blood pressure data is unavailable at the time of prediction. There were eight main objectives pursued in this thesis:

1. To construct valid hourly labels for sustained hypotension under sparse charting conditions, using a window-based persistence proxy that tolerates documentation gaps in MAP observations.

2. To build baseline and advanced models to evaluate under the same locked protocol.

3. To evaluate all models under a fixed 10% alarm budget, where thresholds are calibrated on the validation set and frozen before test evaluation, so that the comparisons reflect performance at equal workload.
4. To report both hourly metrics (AUROC, AUPRC, recall, precision) and event-level metrics (event recall, event miss rate, lead time, cooldown-adjusted alert burden), since event-level outcomes more directly reflect clinical utility.
5. To engineer a broad feature set covering 20 clinical signal blocks (353 columns total) spanning blood pressure, vital signs, laboratory results, vasopressor and sedation infusions, fluid balance, respiratory support, ventilator settings, neurological assessment, and derived metabolic indicators.
6. To conduct systematic ablation studies across two suites totalling 30 model variants, quantifying how much predictive information each signal block contributes and whether non-BP features improve performance on the full cohort.
7. To assess MAP-missing robustness through stratified evaluation and subset-calibrated thresholding, testing whether the alarm system provides more uniform coverage across patients with different monitoring patterns.
8. To demonstrate the MAP-missing coverage disparity (that globally calibrated thresholds produce zero alarms for MAP-missing hours) and to propose and evaluate a mitigation strategy.

## 1.4 Contributions

This thesis makes several contributions to the problem of ICU hypotension early warning under sparse non-waveform data.

At the data level, it introduces a window-based persistence proxy for defining sustained hypotension events on hourly charted data, addressing the failure of strict consecutiveness rules when MAP is charted intermittently. At the feature engineering level, it produces a 353-column hourly panel from MIMIC-IV v3.1 covering 20 clinical signal blocks with standardised temporal feature families, and uses an ablation study across two suites to quantify the per-block signal contribution of each family. Results show that blood pressure-containing models dominate overall performance. The bp_only configuration achieved an AUROC of 0.955 using 37 features, while non-BP variants plateaued around 0.75 AUROC. Adding more features on top of blood pressure did not produce a clear practical improvement and in several configurations slightly degraded performance.

At the evaluation level, the thesis applies a fixed 10% alarm budget with frozen validation thresholds, event-level metrics (event recall, event miss rate, lead time, cooldown-adjusted alert burden), and a MAP-missing stratified evaluation protocol with three calibration regimes. This three-regime approach of global, MAP-missing and MAP-observed, enables performance to be assessed separately for hours with and without blood pressure data, which none of the hypotension studies reviewed for this thesis explicitly report [7].

The core finding is that a globally calibrated alarm threshold produces zero alarms for over half of all patient-hours when MAP is absent, and that this failure is invisible in standard aggregate evaluation. Subset-calibrated thresholds on the same model recover 93.75% recall on MAP-missing hours without any retraining (95% CI: 79.2%–99.2%, $n = 32$ events). Bootstrap resampling confirmed this estimate is stable across 1,000 resamples, indicating that the remaining uncertainty is attributable to the small event count rather than threshold instability. The recommended deployment configuration model (Config G) uses a single lightweight tree model (16 to 37 features, no GPU required) with two thresholds conditioned on MAP availability, achieving 97.6% overall event recall at 0.89 alerts per 24 hours. This outperforms all gated two-model alternatives while requiring no specialised hardware infrastructure, making it operationally simple enough for clinical settings where continuous arterial monitoring is unavailable, though prospective multi-site validation would be required before deployment.

## 1.5 Thesis Report Structure

The remainder of this thesis is organised as follows.

Chapter 2 (Background and Related Works) establishes the clinical, operational, and data-level context. It introduces sustained hypotension as a deterioration signal, frames early warning as an alarm policy problem, reviews prior hypotension prediction work and examines the specific data realities of MIMIC-IV non-waveform charting.

Chapter 3 (Methods) describes the full study design including feature construction, model architectures (logistic regression, HGB, XGBoost), the ablation study design, and the evaluation framework.

Chapter 4 (Experiments and Results) presents all experimental findings, including baseline, ablation results, gated policy evaluation and error analysis.

Chapter 5 (Discussion) interprets the results, and covers methodological implications, limitations and directions for future work.

Chapter 6 (Conclusion) summarises the thesis, restates the key findings, and provides practical recommendations for building early warning systems in similar settings.

# 2 Background and Related Works

This chapter establishes the clinical, operational, and data-level context that motivates the design of the early warning system developed in this thesis. It introduces sustained hypotension as a deterioration signal in intensive care and frames early warning as an alarm policy problem, where models must be evaluated under explicit alert burden constraints rather than ranking metrics alone. This chapter reviews prior work on hypotension prediction and showcases how differences in event definitions, especially between waveform and non-waveform studies, limit study comparability and shape the prediction task itself. The final section examines the specific data realities of MIMIC-IV non-waveform charting, including sparse MAP observations, the failure of strict consecutiveness rules under irregular documentation of signals, and the case for treating missingness and measurement timing as informative features rather than noise in the data.

## 2.1 Clinical and Operational Context

Mean Arterial Pressure (MAP) is a key physiological signal used in ICU settings. It represents the average arterial pressure over a single cardiac cycle. In critical care, clinicians do not only care whether a patient's blood pressure is low at a single moment, but whether the low blood pressure is sustained and/or persistent, increasing the risk of further harm. Clinical discussions of hypotension therefore combine two elements: a threshold (a MAP value below which pressure is considered dangerously low) and a persistence requirement (how long pressure must stay below that threshold before the episode warrants intervention). For example, in cases of septic shock, international guidelines recommend an initial MAP target of 65 mmHg for adults who require vasopressors, reflecting that this is a commonly used threshold for low MAP [3]. This does not mean that this MAP threshold is "safe" for all patients, but is a pragmatic target supported by clinical reasoning and trial evidence that higher MAP thresholds do not yield better results in unselected populations of septic shock [8].

Many observational studies suggest that hypotension exposure leads to worse outcomes in ICU populations. ICU studies that quantify hypotension burden report associations between hypotension and outcomes such as in-hospital mortality and acute kidney damage [1]. More recently, evidence points that hypotension during ICU stay is associated with increased mortality in the majority of included studies, with associations often increasing as hypotension worsens [2]. These findings support the clinical intuition that sustained hypotension is linked to illness severity, treatment decisions and measurement practices, which shows that sustained hypotension is a meaningful deterioration signal and a plausible target for early warning systems. Therefore, the motivation for prediction is not just to classify "low-MAP" hours, but to identify high-risk periods before sustained hypotension develops, when clinicians could plausibly re-assess and intervene.

### 2.1.1 Early Warning as an Alarm Policy

In ICU settings, predictive models are only useful if they can be converted into a more manageable alerting strategy. A model that is outputting a risk score every hour (or more) must decide when to interrupt clinicians. The decision to alert a clinician has a cost: Alerts compete with other important tasks for limited time and attention. This operational constraint is not hypothetical; hospitals routinely report high volumes of device alarms, many of which are non-actionable, and alarm overload has been formally recognized as a patient-safety problem. The Joint Commission's Sentinel Event Alert on medical device alarm safety highlighted that failure to respond to alarms and alarm related adverse events is a recurring safety issue [4]. In addition to severe events, the day to day impact of alarm overload is that frequent, low-value alarms can reduce trust, increase cognitive load, and promote workarounds [9].

A systematic review focused on intensive care nursing reported that alarms are perceived as burdensome and frequent, interfere with patient care, and contribute to reduced trust in existing alarm systems [5]. Although precise "non-actionable alarm" rates vary by device, unit, and definition, the repeated observation across settings is that high sensitivity without selectivity creates noise, and noise can degrade safety by eroding attention and responsiveness. This also has an implication for machine learning related early warning systems, which is that improving ranking metrics (e.g. AUROC) is not sufficient if the operating threshold produces an unacceptable alert volume. Conversely, even slight changes in model performance can matter if they reduce missed events at a fixed acceptable alert burden.

Henceforth, it is useful to view early warning as an alarm policy design problem. Managing alarms safely in clinical settings needs explicit objectives, measured alarm burden and continuous improvement to reduce non-actionable alarms [10]. From this perspective, a machine-learning model is one component of a broader alarm system; the model suggests a score but the thresholding and alerting logic determine how often clinicians are interrupted and how the alarm behaves.

### 2.1.2 Fixed Alarm Budget

When models are evaluated solely as classifiers, results are usually presented using threshold-free ranking metrics (AUROC, AUPRC). These metrics are important for comparing discrimination, especially under class imbalances, as these metrics summarise discrimination by integrating performance across all possible decision thresholds, which is especially useful when the positive class is rare and any single threshold can give a misleading picture [11, 12]. However they do not answer the core question of "how many alarms will the system generate?". In a clinical environment, thresholds imply workload. Therefore, an evaluation that does not explicitly control alert rate risks sacrificing its utility. A model can achieve a high AUROC yet be unusable if it requires excessive alerting to achieve acceptable sensitivity, or if it produces frequent non-actionable alerts.

Therefore, a fixed budget view makes the constraints explicit, which is to choose an operating point that corresponds to a realistic alert rate, then compare models under the same burden. This aligns with how alarms are managed in practice, with thresholds, delays and priorities being tuned to balance sensitivity against alarm load [10]. In studies, a principled approach is to select a threshold on validation data to achieve a target alert rate (e.g. 10% of patient-hours), then freeze that threshold and report test performance at that burden [13]. This approach makes tradeoffs comparable where we can ask "At the same alarm budget, which model misses fewer events, and what lead time is achieved?"

Early warning performance is not fully captured by hour-level accuracy. Clinicians typically need at least one timely warning before a sustained hypotension event, rather than repeating alarms for the same episode. Therefore, event-level outcomes (i.e. such as whether an event is detected at least once in the pre-onset window) is an important metric. In an alarm-system, the central goal becomes to get the event miss rate under a fixed budget and to gauge the alert burden under a realistic alert logic [4].

### 2.1.3 Summary

In conclusion, sustained hypotension is a clinically meaningful deterioration signal, MAP thresholds around 65 mmHg are widely used operationally in critical care settings, and hypotension exposure has been repeatedly associated with adverse outcomes in ICU settings [1–3, 8]. At the same time, ICU environments are alarm dense, with alarm fatigue as a recognised safety problem [4, 5]. These issues motivate an evaluation approach that requires early warning models to be compared at a fixed alert burden, with an emphasis on missed events and alert volume under plausible alert logic [10]. This thesis uses that framing in its study, focusing on the ability to provide timely warnings while explicitly controlling alarm rate.

## 2.2 Prior Work on Hypotension Prediction and Event Definitions

### 2.2.1 Hypotension Prediction

Hypotension prediction has been studied across many medical settings, but the literature is spread across different types of data and changes depending on monitoring conditions. The two main divisions of work are either based on continuous waveform signals (especially arterial line waveforms) and work based on EHR charting data (periodically recorded vital signs). This distinction affects the model design, how hypotension events itself are defined and how prediction performance should be evaluated.

In studies based on waveform-based data, the input data is dense and continuous and is recorded at a second- or minute-level resolution, This makes it easier to define a hypotension event by just using a simple persistence threshold rule (i.e: MAP below 65mmHg for 15 minutes ). For example, the PhysioNet/Computing in Cardiology 2009 challenge on acute hypotensive episode prediction, which used minute-level MAP and persistence based event definition [14, 15].

A large amount of literature on the subject, especially in anaesthesia and perioperative monitoring, uses this waveform data approach. Many recent studies use arterial pressure waveform features and machine learning models that predict impending hypotension, including work around the Hypotension Prediction Index and related systems [16, 17]. These studies often report strong predictive performance but that is due to them having denser data than ICU EHR charting, which means that their event definitions and performance metrics are often built around continuous monitoring and minute level observations that do

not directly translate into a non-waveform model data.
In contrast, ICU studies using EHR sources (e.g. MIMIC data) typically work with irregularly observed vital signs, measurements and heterogeneous recording frequencies. In this setting, this means that a large task is to be able to extract and learn from sparse measurements and data and discover recent trends [18–20]. This setting aligns more with this dissertation, wherein prediction is performed on an hourly grid using non-waveform MIMIC-IV v3.1 data [6]. In this context, the key challenge is not just to predict hypotension but to also be able to extract meaningful signals from sparse and irregular data.
This difference is important when measuring results as a model that performs well on minute-level forecasting with arterial waveform data is not directly comparable to a model that predicts a risk score over hourly data using charted EHR data as the signal density, event rarity and label definitions of the problem are different. Due to this, this thesis keeps the task framing and does not make direct comparisons across studies that use fundamentally different data resolutions and definitions.

### 2.2.2 Difference in Event Definitions

Hypotension literature has varying definitions of what is considered a hypotension event, even when studies are tackling the same clinical problem. Most studies do use a MAP threshold but they differ in threshold value, required persistence and the resolution of time of the data.

However, a MAP threshold of around 65mmHg is widely used in critical care practice because it is clinically meaningful and appears in guideline-based discussions of the subject [3]. However, it is to be noted that some studies use a lower threshold (such as 60mmHg), mostly in benchmark related studies or others by combining lower thresholds with different duration constraints [14–16].

For work based on waveform data, persistence is usually defined in several sustained minutes or majority below threshold values for a certain time window. However in chart-based EHR work, persistence has to be implemented differently based on the available sampling density [20, 21].

For contrast, the PhysioNet AHE challenge illustrates how waveform-native definitions break down in this setting. That challenge used a minute-level persistence criterion: the majority of MAP values had to fall below the threshold over a prolonged interval [14, 15]. With dense waveform data, that rule works because dozens of MAP readings are available within each window. In hourly EHR charted data, the same logic cannot be applied directly because there may be no MAP value recorded for several consecutive hours.

This variance in definition shapes the task. A more relaxed definition generates more events and causes an increase in the apparent recall, while a stricter definition would reduce false positive but may also reduce events and make prediction tougher. Similarly, changing the time resolution also changes what "persistence" classifies as. Due to this, performance numbers in studies are dependent on event definition and time resolution in context [7].

### 2.2.3 Definition Sensitivity in Non-Waveform Data

As the previous section shows, two studies claiming to predict 'hypotension' may differ in what counts as an event, when it begins, and what evidence confirms it [7, 16]. This is particularly important in non-waveform ICU datasets, where the data is irregular and event definitions must be adapted to charted data rather than continuous signals. This means, while two studies may claim to predict "hypotension", they may differ substantially in what counts as an event, when the event is considered to begin and what evidence is needed to confirm it [7, 16].

Definition sensitivity is defined by three choices, those being threshold, persistence and time resolution. Thresholding choice is the most visible (e.g MAP $<$ 65mmHg versus MAP $<$ 60 mmHg), but it is often not the reason for variation as despite determining the level of MAP drop required to consider a hypotension event, it is mostly set around the 65mmHg value. Persistence rules have a large effect, as they determine whether a temporary drop in value is treated as a hypotension event or excluded as a short term instability [14, 15]. Time resolution then interacts with persistence, therefore a rule that works at a minute-level waveform resolution may not be viable when applied to hourly values. As a result, the same physiological concept can produce different labels depending on the definition of the event.

This is important as label definition changes the structure of the task itself. Consequently, these changes affect not only AUROC and AUPRC but also the precision, recall and alarm budget as well [20, 21]. In other words, changes in event definition change the class distribution and difficulty of the prediction problem at its core level.

In non-waveform ICU studies, event definition must be built on observed charted values, not continuously monitored vital data. This leads to ambiguity about what would happen when no measurement is recorded at a candidate time point. Some studies avoid this issue by using shorter observation time windows, different aggregation rules or a more relaxed criteria, while others rely on assumptions of the data [18, 19]. Thus, "non-waveform hypotension prediction" is not a standardized task but a family of related tasks depending on the mapping of irregular data into its definition labels.

A consequence of this issue is that cross-study comparisons can be quite misleading when the details of the event definition are not taken into account. For example, two models evaluated on different hypotension definitions may report similar AUROC values while operating with different event definitions, occurrences and onset structuring. Therefore, a model that may apparently perform worse under one definition may perform better under a different event definition simply due to the label matching the temporal pattern captured by model features better. This is quite relevant in retrospective ICU studies where feature engineering, missingness handling and label construction are intertwined [7, 22]. Due to this reason, any meaningful comparison requires context on how the event was defined and the context of the setup.

Therefore, from a design perspective, the variability of definition should be chosen to match the data and its intended use, rather than being taken from an entirely different monitoring context. In waveform models, strict duration-based hypotension persistence can be justified because many observations are available over short time windows [14, 15]. In non-waveform ICU charting, the same logic may need to be reformulated to preserve the intended clinical meaning under sparse and irregular data. This is not a deviation from previous work but an adaptation of the core idea of hypotension prediction targeting sustained instability of hemodynamic features (Arterial and Blood Pressure features) and not isolated instability incidents.

Hence, the main challenge in the thesis is not just to find which model predicts the best but also which event definitions remain interpretable and usable in a sparse non-waveform hourly signal.

## 2.3 Data Reality in MIMIC-IV Non-Waveform Data and Design Implications

### 2.3.1 Hourly Grids and Irregular Charting

The core modeling dataset created for this thesis contains hourly data. The patient state is represented on an hourly grid over the first 48 hours of their ICU stay. This choice matches the intended use of the model as an early warning system. In MIMIC-IV, from which this thesis's dataset was derived, the underlying source data is not generated as a clean, physiological stream with regularly charted data. Rather, it is created through routine clinical documentation, bedside charting, data from different devices and administrative records, all of which occur at irregular times across variables and patients [6, 23]. Therefore, the hourly grids used for modeling are not native to the dataset, but a representation created after careful processing, and which took the greatest effort in this thesis.

This distinction matters because existence of an hourly row for a patient does not necessarily mean the row contains all clinically relevant variables for that hour, as they may not have been measured during that hour. Some variables may have been recorded at different times or not at all. Therefore, the absence of a value in a given hour cannot be interpreted as evidence that the physiological state was normal or unchanged. In EHR-based deterioration monitoring, **not measured** and **normal** are not equivalent. This is supported by the fact that documentation studies show that vital sign recording is shaped by hospital workflows, staff behaviour and local practices, meaning that data availability is a reflection of the care process rather than being a uniform observation protocol [24]. Thus, it is unsafe to interpret missing measurements as observations.

The usage of time bins is relatively common; benchmarking pipelines from MIMIC data routinely resample irregular time series data into one-hour intervals as this enables repeated prediction, simpler feature construction and more reproducible comparisons [25–27]. However, these pipelines also specify that "hourly data" in critical care machine learning is usually a representation of charted events and not a fully observed continuous signal. Due to this, the hourly grid in this thesis should be regarded as a decision framework which defines the times at which the model is allowed to issue an updated risk score. It does not remove the irregularity from the underlying data but instead uses the sparse data to predict the risk score with the variables it has.

This is especially relevant for blood pressure variables. In a waveform setting, it is safe to assume that MAP values will be available almost continuously once an arterial line is present. However, in non-waveform charting, MAP may appear only intermittently in the data, may have to be derived from other blood pressure vitals or may be absent for long periods of time. Therefore, the hourly representation is a clinically meaningful but incomplete representation of the patient state. Thus, a modeling strategy is needed that takes into account what was actually observed and does not assume continuity of data.

### 2.3.2 Sparse MAP Observations and Consecutiveness

After the data is organised into hourly rows, the first assumption might be to define sustained hypotension using strict consecutiveness (requiring hypotension events in consecutive hours), in order to mirror clinical intuition that sustained hypotension should persist over time rather than being an isolated threshold as discussed. In dense waveform data, such duration-based rules are straightforward to implement because a large number of observations are possible within each time window [14, 15]. However, in non-waveform EHR data this logic becomes difficult to hold.

The main issue is that consecutive hourly evidence requires both physiological and observation density. However, a patient may plausibly remain hypotensive for several hours, but if MAP is not recorded for an hour, a strict consecutive low MAP based rule would treat the sequence as interrupted. This means that the rule is not only testing whether hypotension persisted, but also whether documentation happened to occur frequently enough to confirm persistence on the imposed hourly grid. In sparse charting, the latter dominates the former. As a result, event construction becomes sensitive to documentation density which is not the case in dense waveform data.

This is not simply a preprocessing inconvenience but directly affects the validity of the target label. If the persistence rule is too strict in relation to the observation process, then clinically valid hypotension episodes may not get detected in the dataset or may have their onset shifted to a later hour where documentation has more signal points. Conversely, if persistence is ignored in its entirety, then even isolated low MAP values may be treated as a hypotensive episode. In both cases, the event definition would no longer be considered a neutral technical detail as it shapes the prediction task itself [7, 20].

The literature on ICU data quality supports the broader concern that even vital physiological signals can have incomplete temporal coverage in critical care datasets. Analysis of MIMIC vital-sign time series have shown that completeness and continuity vary substantially across variables and stays, particularly when going from raw monitor outputs to structured analytic grids [28]. This supports an important methodological point in this thesis, that in non-waveform hourly data, the challenge is not just to predict hypotension but to also define persistence rules that remain meaningful under sparse data.

Due to this, a persistence rule for non-waveform data cannot depend completely on uninterrupted hour to hour observation. It must tolerate documentation gaps while still requiring low blood pressure signals. How this thesis constructs this rule is described in Section 3.4.2.

### 2.3.3 Missingness and Measurement Timing as Signals

Another important data consideration is that missingness in ICU EHR data is often more informative than random. Whether a variable is measured, how often it gets measured and how recently it was last observed are a reflection of the patient's acuity, nursing workload, clinical concern given and local monitoring practices. This information has been formalised in this literature as being an Informative Observation, where the observation process itself carries predictive information about outcome risk [29]. In other words, the measurement schedule is a part of the data itself.

This is quite relevant in an ICU early warning setting. A patient who is being measured frequently may be someone for whom clinicians already suspect instability in vitals, and long gaps in measurements may occur during relatively stable periods or may simply be a different care context. The presence of data is therefore not independent of the patient's condition. This makes observation timing potentially useful for prediction, especially in retrospective EHR studies where the goal is to model what could have been inferred from the information available at each time point.

Prior work in clinical time-series modeling has incorporated this insight in different ways. Some approaches explicitly provide masks showing whether each variable is observed at a given time, while others model time gaps or use decay on previous observations based on time [22, 25]. The central idea is that the model should not be forced to treat observed and unobserved values as being equivalent but should treat it as a feature itself.

At the same time, informative observation should be interpreted carefully. Measurement frequencies may be able to capture clinical concern, but they can also reflect workflow differences, staffing, or healthcare-process bias. Work on EHR Bias shows that aspects of the care process, including when tests are ordered or data is recorded, can itself become strong predictors [30]. This does not invalidate its use, but it means that they are represented transparently rather than being hidden through preprocessing steps that erase the distinction between observed and unobserved periods. In this thesis, that leads to the use of **observation-pattern features**, such as observation flags, recency measures, and recent measurement counts, as part of the feature space itself. Section 3.5.8 describes their implementation.

### 2.3.4 Why Forward Fill Is Avoided

Given the sparse hourly charting of data, a common preprocessing practice is to impute the missing values so that every row has a complete feature vector. One of the most common approaches is to carry the last observation forward, or **forward fill**, where a missing value is replaced by the most recent past observation. This is also embedded in several common benchmarks preprocessing pipelines for ICU prediction tasks [25, 26]. However, the convenience of forward fill does not imply its neutrality.

Statistical literature has repeatedly noted that last observation being carried forward in methods can introduce bias because they assume, implicitly or explicitly, that the most recent observed value remains a reasonable substitute for the unobserved current value [31]. In clinical data, that assumption is hard to justify as the patient state may change rapidly and the missingness may itself be related to the underlying trajectory. More importantly, forward fill is problematic when the target definition itself depends on the temporal persistence in an observed physiological variable.

If a hypotension episode is defined using repeated low MAP evidence over time, then carrying a low MAP value forward across unmeasured hours risks creating **artificial persistence**. This can make an episode appear more continuous than the actual evidence even supports, can shift onset earlier or later depending on the pattern used for imputation, and can alter which hours are labeled positive. Thus, forward fill does not just complete the feature matrix, it can change the target itself. That is a much stronger effect than ordinary feature imputation, because it affects the meaning of the outcome itself.

Therefore, in this thesis, missingness is not ignored, and is instead represented through observation pattern features and recency features. This approach is more consistent with missingness-aware modeling frameworks, where the model is informed about what was seen and when, rather than giving imputed data [22, 29]. Section 3.5.8 describes the specific observation-pattern features used in this thesis.

### 2.3.5 Summary

The previous sections establish several constraints that shape the design of the prediction system in this thesis. Sustained hypotension is clinically meaningful but must be defined carefully in non-waveform data, where sparse MAP observations prevent strict consecutiveness rules from reliably identifying episodes. The hourly grid is a useful decision framework but does not guarantee complete observation, so the absence of a value cannot be equated with normality. Missingness and timing carry information about the patient and clinical attention, making them appropriate model inputs rather than being data to be erased in preprocessing. Forward fill of vital signs is avoided because it risks creating artificial persistence in the very signal used to define the target label. A limited exception is made for laboratory results, where a causal last-known-value feature carries only past observations forward without crossing the onset boundary (Section 3.5.8). This exception does not affect the target definition because laboratory values are not used to construct the hypotension label.

These considerations lead directly to the methodological choices described in Chapter 3: a window-based persistence proxy that tolerates charting gaps, a valid mask that excludes all post-onset hours (Section 3.4.4), observation-pattern features that encode what was measured and when (Section 3.5.8) and a fixed alarm budget evaluation that controls alert burden rather than relying solely on ranking metrics (Section 2.1.2). Each of these decisions is a response to the data realities and clinical constraints outlined in the above chapter, their implementation details follow in Chapter 3.

# 3 Methods

## 3.1 Study Design Overview

This study is a retrospective observational modeling study using EHR (Electronic Health Record) data from the MIMIC-IV v3.1 (Medical Information Mart for Intensive Care, Version 3.1) [6]. The objective is to develop and evaluate an hourly early warning system for detecting sustained hypotension events in adult ICU patients, which has been framed as an alarm policy problem under a given fixed alert budget. This study follows the development and validation design, consistent with the TRIPOD framework for transparent reporting of clinical prediction models [32].

The prediction task is defined for individual patient hours. For each ICU stay, the model outputs a risk score every hour on a temporal grid, and these scores are converted into binary alarms using a threshold calibrated to a target alarm rate on the validation set. This threshold is then frozen and applied to the test set. Then the evaluation reports both the hourly metrics and the event-level outcomes under the fixed alarm budget, which reflects the operational reality that clinicians would require at least one warning before a hypotension event rather than a continuous per-hour classification accuracy [10, 13].

The research question is whether an early warning model can maintain clinically useful performance when Mean Arterial Pressure (and related blood pressure signals) is missing from the input data. This question arises because MAP, which is also the dominant predictive signal, is absent in approximately 56% of patient hours in the dataset due to irregular charting practices. Standard evaluation on the full cohort can mask a deployment failure that only becomes visible when results are broken down by MAP availability: when a single alarm threshold is calibrated on the combined population, the model can produce near-zero alarms for patients without recent MAP observations. Put simply, roughly half of all patient hours become invisible to the alarm system, not because those patients are low risk, but because their risk scores fall in a range that a globally calibrated threshold was never designed to reach.

This score distribution shift is the core observation that motivates the evaluation strategy in this thesis. When MAP is available, blood pressure features produce high, well separated risk scores. When MAP is absent, the model relies on weaker indirect signals and its scores cluster in a narrower, lower range. A threshold set on the full validation cohort is dominated by the MAP observed majority and ends up too high for the MAP missing subset, so patients in that subset receive no warnings at all. Recognising this, the study evaluates all models under three calibration regimes: a global threshold calibrated on the full validation cohort, a MAP missing calibrated threshold tuned specifically to the subset of hours without MAP observations, and a MAP observed calibrated threshold tuned to hours with MAP observations. Comparing performance across these regimes separates genuine discriminative ability from threshold artifacts caused by the distribution mismatch between subsets.

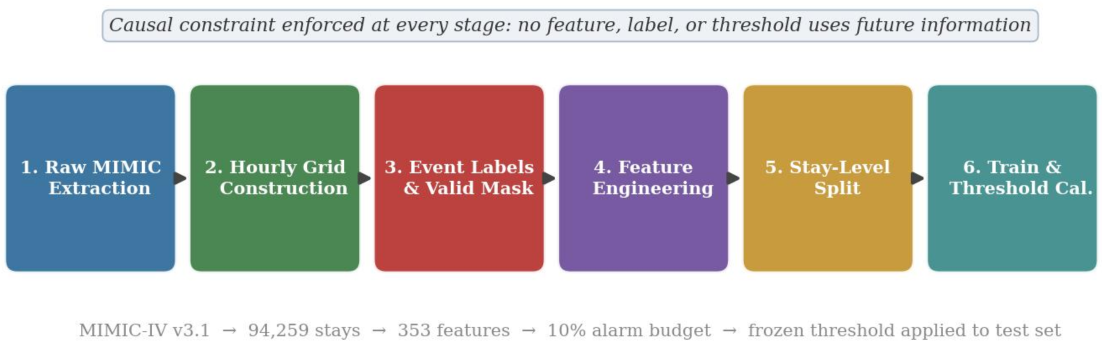


Figure 3.1: Modeling pipeline overview. Each stage enforces causal constraints so that no feature, label, or threshold uses information from future hours.

The modeling pipeline has six stages:

1. Raw MIMIC extraction.
2. Hourly Grid Construction
3. Event Labeling and Validity Masking
4. Feature Engineering
5. Stay-level data splitting
6. Model Training with Threshold Calibration

Each stage of the pipeline enforces strict constraints to prevent data leakage of future hours to current hour predictions.

At a high level, the study proceeds as follows. Clinical data from MIMIC-IV is organised into an hourly grid, labeled with a sustained hypotension target, and split by patient into training, validation and test partitions. A set of 353 features spanning blood pressure, vital signs, laboratory results, interventions, respiratory support and demographics is engineered from the raw records. Three model architectures are trained (a logistic regression baseline and two gradient boosted tree ensembles), and each model's alarm threshold is calibrated on the validation set to produce a fixed 10 percent alarm rate. Performance is then evaluated not only on the full test cohort but separately on the subset of hours where MAP was observed and the subset where it was missing, which is the stratification that reveals the deployment failure central to this thesis. Finally, an ablation study systematically adds and removes feature groups to measure which clinical signals contribute useful predictive information, with particular attention to whether non-blood-pressure signals can sustain performance when MAP is unavailable.

The design choices for each stage are further discussed below.

## 3.2 Cohort and Data Source

The data source used is MIMIC-IV v3.1, a de-identified electronic health record dataset that covers patients admitted to the Beth Israel Deaconess Medical Center between 2008 and 2022 [6]. The dataset is hosted on PhysioNet [33] and access to the dataset was obtained by completing PhysioNet's credentialed data access program, which requires completing training courses on data usage and subsequent data usage agreement. Since the data is fully de-identified in accordance with the Health Insurance Portability and Accountability Act, the dataset qualifies for a waiver of informed consent for secondary research use.

The study uses the ICU module and selected tables from the hospital module of the data. From the ICU module, `icustays` provides the mapping between the hospital admissions and the individual ICU stays, including the timestamps for admission and discharge timestamps that anchor all temporal features. `Chartevents` contains bedside-charted clinical observations such as vital signs, ventilator settings and neurological assessments, with item identifiers being resolved through `d_items`. `Inputevents` records medication administrations including vasopressor and sedation infusions with start times, end times and rates. `Outputevents` records fluid outputs, including urine. `Procedureevents` provides records of procedures such as renal replacement therapy. From the hospital module, `labevents` contains laboratory results timestamped to the time of specimen collection, with item identifiers being resolved via `d_labitems`. Prescriptions provides medication order data that are used for antibiotic exposure proxies and `diagnoses_icd` provides discharge diagnoses codes used for comorbidity prediction. Not all fields from these tables are used.

From `chartevents`, only specific item identifiers are extracted: MAP, SBP, DBP, heart rate, SpO2, respiratory rate, temperature, FiO2, oxygen flow rate, ventilation mode, PEEP, tidal volume, minute ventilation, and GCS total. From `inputevents`, five vasopressor agents (norepinephrine, epinephrine, phenylephrine, dopamine, vasopressin), five sedation agents (propofol, midazolam, dexmedetomidine, fentanyl, morphine), and all intravenous fluid records are extracted. From `outputevents`, only urine volume is used. From `procedureevents`, only renal replacement therapy events are used. From `labevents`, thirteen analytes are selected (lactate, creatinine, BUN, sodium, potassium, chloride, bicarbonate, glucose, haemoglobin, WBC, platelets, total bilirubin, pH), chosen because they reflect organ function, metabolic state, or infection severity relevant to hemodynamic instability. From `prescriptions`, only antibiotic orders are extracted. From `diagnoses_icd`, only the codes needed to derive a Charlson Comorbidity Index and four binary comorbidity flags are used. All other fields and tables in MIMIC-IV are excluded.

This study cohort includes all adult ICU stays in MIMIC-IV v3.1. The observation window is restricted to the first 48 hours of each ICU stay; this is a common choice in ICU prediction studies that balances sufficient observation time against confounding from prolonged stays wherein the care patterns and patient trajectories become increasingly heterogeneous [25, 34]. This has 94,259 unique ICU stays and 4,524,432 (about 4.5 million) total patient-hour rows. After applying the valid mask, which excludes post-onset hours (where hypotension has already begun and prediction is no longer early warning) and hours 0 through 5 (which serve as a baseline window during which patient specific reference values such as median MAP are computed, so that later features can measure deviation from each patient's own normal), 3,590,802 (about 3.6 million) rows remain eligible for prediction. Of these 19,520 carry a positive early warning label, which gives us a valid row prevalence of 0.54 percent. This severe class imbalance is characteristic of hourly ICU prediction tasks and motivates the fixed alarm budget evaluation framework [12].

Only non-waveform charted data is used; all physiological signals are derived from structured clinical documentation rather than continuous waveform monitors, as discussed in Section 2.3.1. This choice reflects the intended context of deployment: a system that operates on routinely available EHR data without requiring specialized waveform infrastructure. A lightweight model can run on charting data on hospital monitoring and reporting systems. It also means that all signals are subject to the documentation irregularities, variable measurement frequencies, and informative missingness patterns described in Section 2.3.3 [24, 29].

## 3.3 Time Representation and Hourly Grid

All clinical observations are mapped onto a fixed hourly grid aligned to the ICU admission time. As discussed in Section 2.3.1, the underlying MIMIC-IV data is not recorded on a regular schedule but arises from routine clinical documentation at irregular times, so this grid is an imposed representation rather than a native property of the data. For each event recorded in MIMIC-IV, the ICU relative hour is computed as the difference between the event timestamp and the ICU admission time (the time from `icustays`, divided by 3600 seconds). Only hours in the range of 0 to 47 inclusive are kept, which corresponds to the first 48 hours of each ICU stay. The resulting dataset has one row per unique combination of stay identifier and hour, and this pair serves as the primary key for all feature construction, labeling and evaluation.

This hourly binning strategy is standard in ICU prediction pipelines built on MIMIC data. Harutyunyan et al. [25] established an influential benchmark that resamples irregular clinical time series into a fixed one-hour interval for mortality, length of stay, and phenotyping tasks. Wang et al. adopted the same approach in their MIMIC Extract preprocessing pipeline [26], and Puroshotham et al. similarly discretise irregular clinical events into fixed time bins for benchmarking deep learning against traditional machine learning and clinical severity scores [34]. The hourly resolution represents a practical compromise which is fine enough to capture clinically meaningful changes in patient state, coarse enough to tolerate the irregular charting frequencies of the non-waveform ICU data, and directly interpretable as a decision point at which the early warning system can issue an updated risk assessment.

It is important to recognise that an hourly row does not guarantee that all variables were observed during that mentioned hour. A patient may have vital signs charted irregularly and not be present for the aforementioned hour, and laboratory results may be charted only every several hours. Therefore, many cells in the feature matrix are also missing by design, reflecting the clinical decisions about monitoring intensity and test ordering as discussed in Section 2.3.3 [29, 30]. Rather than imputing these gaps or resorting to forward fill of vital signs, the pipeline instead preserves missingness and encodes it explicitly through observation pattern features. For each signal, binary observation flags, hours since last observed counters, and within window measurement timing is informative rather than being noise, which is consistent with frameworks for informative observation in clinical time series [22]. The only exception is laboratory values, where forward fill is applied to produce a last known value feature, hence the most recent past result is carried forward to represent a clinician's knowledge of that current hour. This forward fill method uses only past observations and does not cross the onset boundary.

Within each hourly bin, multiple observations of the same variable are then aggregated. For continuous signals such as vital signs, the aggregation then takes the mean of all observations falling within that hour. For binary event signals such as vasopressor activity or ventilator status, the aggregation would then produce a flag indicating whether the event was active at any point during the hour. Interval-based records such as medication infusions, the pipeline expands each infusion interval into the set of hours it overlaps and marks each overlapping hour as active. This expansion uses only the floor and ceiling of the start and end times respectively, ensuring that an infusion starting partway through an hour is attributed to that hour but not to any future hour. This within-hour aggregation and interval expansion logic is applied consistently across all signal types and enforces causal constraint that no information from beyond the current hour is available to the model at prediction time.

## 3.4 Hypotension Event Definition

### 3.4.1 MAP Threshold and Event Construction

The prediction target is the onset of sustained hypotension, defined by using Mean Arterial Pressure. MAP values are taken from `chartevents` using item identifiers resolved via the MIMIC-IV item dictionary. MAP is sometimes recorded directly (for example, from arterial line monitors) and sometimes not charted at all. When MAP is absent, it is not derived from systolic and diastolic blood pressure. The pipeline uses only directly recorded MAP values to avoid introducing estimation error into the variable that defines the target label.

The MAP threshold used is 65mmHg, which is consistent with the widely used clinical target recommended in the Surviving Sepsis Campaign guidelines [3]. An individual hour is marked as being hypotensive if the mean of MAP observations within that hour fall below 65mmHg. Hours without any MAP observations are not marked as being hypotensive regardless of prior values. This means that the event definition is conservative with respect to missingness: unobserved hours cannot contribute evidence of hypotension, which avoids the artificial persistence problem as described in 2.3.4.

### 3.4.2 Sustained Hypotension Persistence Proxy

A single hour of low MAP does not constitute a sustained hypotension event. To capture sustained hypotension events rather than the fluctuations in MAP, the pipeline uses a window based persistence proxy. This requires at least two hypotensive hours within a rolling three-hour window, and additionally requires that at least two MAP observations exist within that window. These conditions ensure that the rule tests both physiologic feature persistence and observational evidence, instead of triggering on one isolated measurement like basic alarms do.

This design is motivated due to the failure of strict consecutiveness in non-waveform data. As discussed in 2.3.2, MAP is absent in approximately 56% of patient hours. Requiring hypotension in consecutive hours would depend on documentation density rather than the physiological state, and in practice yields near zero event stays when it is applied to the MIMIC-IV hourly grid. The rolling window relaxes the consecutiveness rule while preserving the core idea of sustained hypotension requiring repeated low MAP evidence over a short interval of time.

Conceptually, this rule is the hourly operationalisation of the persistence logic used in the PhysioNet Acute Hypotensive Episode (AHE) challenge, which defines an event by MAP being below the threshold for the majority of a time window at minute-level resolution [14, 15]. The thesis rule adapts this principle to hourly resolution window and sparse charted data; instead of requiring 90% of minute resolution readings below threshold over 30 minutes, it requires two out of three hourly slots to contain below threshold MAP evidence. The threshold of 65mmHg is retained from the clinical standard [3].

The onset hour is defined as the first hour at which the persistence proxy becomes true. This is the prediction target, the onset hour before which the model must raise an alarm.

### 3.4.3 Horizon Labeling

The early warning label is constructed by using a fixed prediction horizon of three hours. For each eligible hour `t`, the binary label is set to 1 if a hypotension onset exists in the interval [`t`,`t` + 3], and 0 otherwise. This means the model is asked whether sustained hypotension will begin within the next one to three hours, providing clinicians a plausible window for intervention.

The three hour horizon is a balance of prediction difficulty against clinical utility. A shorter horizon would make the prediction task easier but leave less time for a response while a longer horizon would increase the lead time but reduce the label precision, as more changes would occur in patient state would occur between prediction and event. The same horizon is applied uniformly across all models to ensure comparability.

### 3.4.4 Valid Masking and Leakage Guards

Several constraints are applied to prevent information leakage and ensure that the evaluation reflects genuine early warning capability. First, all hours at or after the onset hour are marked as invalid and excluded from both training and evaluation. Once sustained hypotension has begun, predicting it is no longer early warning but contemporaneous detection, and including these hours would inflate performance metrics of the system. Second, predictions are only generated for hours where `hours_since_icu` is at least 6. Hours 0 through 5 serve as a baseline window from which patient specific reference values (like the baseline median MAP) are computed as allowing predictions during this window would risk the model using features that have not yet stabilised.

Third, all features are constructed using only data from the current hour or earlier. No feature computation uses any information from future hours, including forward looking rolling windows or features derived from post-prediction data. For interval-based records (like vasopressor infusions) the expansion logic attributes activity only to hours that overlap with the infusion interval; an infusion that begins after the prediction hour is not visible to the model at that hour. These constraints are enforced at the feature construction stage rather than relying on post-hoc filtering.

## 3.5 Feature Engineering

### 3.5.1 Feature Family Design Pattern

A systematic template is applied to transform each clinical signal into a family of engineered features that capture current state, recent history, trend and observation patterns. This template is applied across all signal types, ensuring uniform representation and interpretability. For a continuous numeric signal such as MAP, it consists of the following 11 items: the current-hour aggregated value, the most recent prior value, a patient-specific baseline computed as the median over hours 0 through 5, rolling six-hour summary statistics (mean, minimum, maximum, standard deviation), change from baseline, a binary observation flag indicating whether the signal was measured in the current hour, hours elapsed since the last observation, and a count of observations within the current hour or recent window. For binary event signals such as vasopressor activity, the family consists of four items: a current-hour active flag, a six hour any active flag, a count of active hours within the last six hours, and hours since the last active hour.

This design ensures that even when a signal's current hour value is missing, the model still receives information about recency, trend and measurement pattern of the aforementioned signal. The observation pattern features (flags, counters, hours-since) serve as representations of informative missingness as discussed in 2.3.4 [22, 29]. The full feature set totals to 353 columns.

Table 3.1 summarises these two templates using MAP and vasopressor activity as examples.

### 3.5.2 Core Blood Pressure Features (34 Columns)

The blood pressure block is the foundation of the feature set and the dominant predictor in all experiments. It covers MAP, systolic blood pressure (SBP), and diastolic blood pressure (DBP), each receiving the full 11 column numeric family described in Table 3.1. That would produce 33 columns (11 per signal), but MAP also receives one additional column, `MAP_count_6h`, because MAP observation density over the preceding six hours proved informative during early development; SBP and DBP receive equivalent six-hour counts through their own families, bringing the total to 34. Together these features capture the

Table 3.1: Feature family templates applied to each clinical signal. Continuous signals produce 11 features per signal; binary event signals produce 4.

| Template | Feature | Example (MAP / Vasopressor) |
|---|---|---|
| Continuous (11) | Current-hour value | `MAP` |
| | Most recent prior value | `MAP_last` |
| | Baseline (median hrs 0–5) | `MAP_baseline` |
| | 6h rolling mean | `MAP_mean_6h` |
| | 6h rolling minimum | `MAP_min_6h` |
| | 6h rolling maximum | `MAP_max_6h` |
| | 6h rolling std. deviation | `MAP_std_6h` |
| | Change from baseline | `MAP_delta` |
| | Observation flag | `MAP_observed` |
| | Hours since last observation | `hours_since_MAP` |
| | Current-hour obs. count | `MAP_count_6h` |
| Binary (4) | Current-hour active flag | `pressor_active_1h` |
| | Any active in last 6h | `pressor_any_6h` |
| | Active hours in last 6h | `pressor_hours_in_6h` |
| | Hours since last active | `hours_since_pressor` |

current blood pressure state, the trajectory over the preceding hours, and the monitoring pattern itself.

The importance of measurement-pattern features (i.e `hours_since_MAP` and `MAP_observed`) becomes particularly relevant in the MAP-missing analysis, where features carry a predictive signal even when the current hour value is absent. When combined with the 4 static demographic columns that are always included (Section 3.5.7), the bp_only variant uses 37 features in the compressed15 suite. Under feature simplification (Section 3.6), baseline, delta, and rolling min/max/std columns are dropped, reducing the variant to 16 features in the auto-suite.

### 3.5.3 Extended Vitals: HR, SpO2, RR, Temperature (44 Columns)

Four additional vital signs are extracted from `chartevents`: heart rate, peripheral oxygen saturation, respiratory rate and temperature. Each of these features receives the same 11 features numeric columns as the blood pressure signals, which yields 44 features in total. During extraction, only item identifiers corresponding to actual clinical measurements are included; identifiers for alarm low and alarm high readings are excluded, as these represent device threshold triggers rather than measured values and would contaminate feature values. Physiologically impossible entries are also excluded via range validation (e.g heart rate between 0 and 300, SpO2 between 0 and 100).

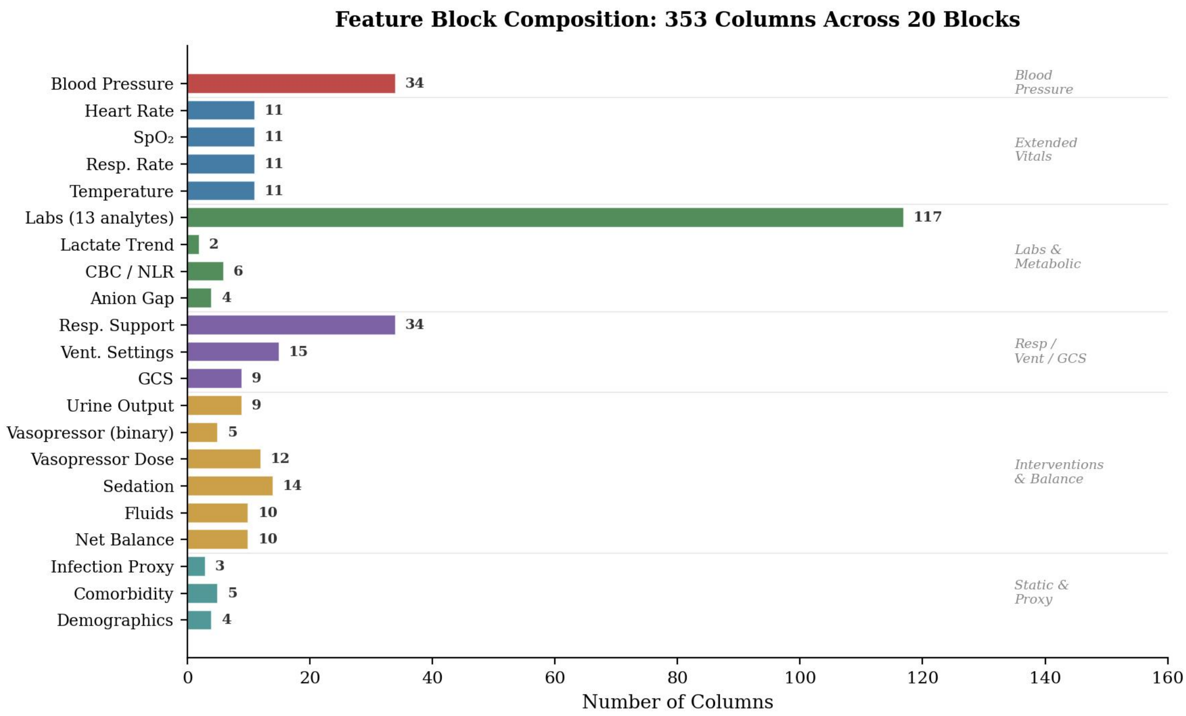


Figure 3.2: Feature block composition. The 353 columns are organised into 20 blocks grouped by clinical domain. The labs block (117 columns) is the largest; blood pressure (34 columns) is the dominant predictor despite its modest size.

### 3.5.4 Urine, Vasopressors, Fluids, Sedation, and Net Balance (50 Columns)

This group of features captures clinical interventions and fluid balance, which are relevant both as indicators of patient state and as potential upstream signals of instability.

Urine signals (9 columns) is derived from `outputevents` and consists of the hourly volume, baseline, six-hour rolling statistics, delta, observation flag, hours-since-last observation, and current-hour observation count. Urine signals are sparsely charted in ICU data, which reflects the nursing practices rather than continuous measurement. The hours-since and observation flag features are therefore particularly important for this block.

Vasopressor binary features (5 columns) are derived from `inputevents` by expanding each infusion interval into the set of hours it overlaps. Four of these follow the binary event template in Table 3.1: a current-hour active flag, a six-hour any-active flag, a count of active hours within the last six hours, and hours since last vasopressor activity. A fifth column, `pressor_count_1h`, records the number of concurrent vasopressor infusions in the current hour, which goes beyond the basic binary template because multiple agents can run simultaneously. In addition to these features, the vasopressor dose intensity features (12 columns) capture per agent infusion rates for norepinephrine, epinephrine, phenylephrine, dopamine, and vasopressin, along with a composite norepinephrine-equivalent dose (NEE).

NEE converts each vasopressor agent into a common scale by multiplying its infusion rate by a conversion factor relative to norepinephrine: norepinephrine and epinephrine use a factor of 1.0, phenylephrine 0.1, dopamine 0.01, and vasopressin 2.5 [35]. The sum gives a single number representing total vasopressor exposure in norepinephrine-equivalent units. NEE provides a single scalar summary of total vasopressor exposure, which is relevant because higher aggregate doses reflect greater hemodynamic support and more severe circulatory compromise.

Sedation and analgesia features (14 columns) capture the exposure of patient to propofol, midazolam, dexmedetomidine, fentanyl, and morphine. For each agent, a binary active flag and hourly infusion rate are recorded, along with aggregate features (any sedation in last six hours, hours since last sedation). These agents contribute to hypotension via direct effect on hemodynamic stability and serve as proxies for severity of illness.

Fluid administration features (10 columns) are extracted from `inputevents` and incude hourly fluid volume, observation flag, baseline, six-hour rolling statistics, delta, and hours-since. Net fluid balance (10 columns) is derived as the difference between fluid input and urine output per hour, with the same rolling and recency features applied to the net balance signal.

### 3.5.5 Laboratory and Derived Metabolic Features (135 Columns)

Laboratory results are extracted from the hospital module's `labevents` table. Thirteen laboratory analytes are included: lactate, creatinine, blood urea nitrogen, sodium, potassium, chloride, bicarbonate, glucose, haemoglobin, white blood cell count, platelets, total bilirubin, and pH. For each analyte, item identifiers are discovered from `d_labitems` using keyword matching and frequency-based selection to identify the most commonly charted variant. Each analyte receives a nine-feature family: current-hour value, observation count, last known value (via causal forward-fill), baseline, six-hour rolling mean, delta, percentage delta, and hours since last observation. This produces 117 columns across the 13 analytes.

In addition to the base laboratory features, several derived metabolic indicators are included. The lactate trend block (2 columns) captures six-hour lactate change and twelve-hour maximum lactate, providing trend information beyond the point value. These columns are non-overlapping with base lactate features in the labs block.

The CBC inflammation block (6 columns) derives neutrophil-to-lymphocyte ratio (NLR) from differential white blood cell counts. NLR has been established as an inflammatory severity marker associated with adverse outcomes in sepsis populations [36]. Features include the most recent NLR value, six-hour rolling mean, observation flag, and hours since last observation.

The anion gap block (4 columns) computes anion gap as sodium minus the sum of chloride and bicarbonate using the most recent available values of each electrolyte. Elevated anion gap reflects metabolic acidosis and has been associated with mortality in ICU cohorts [37]. Features include the most recent value, six-hour long rolling mean, observation flag, and hours since the last observation.

The infection proxy block (3 columns) derives antibiotic exposure indicators from the prescriptions table where available. Features include a current-hour antibiotic-active flag, a new antibiotic start flag, and hours since last antibiotic administration. This block is extracted opportunistically and handled gracefully if the prescriptions table is absent.

The comorbidity static block (5 columns) derives a Charlson Comorbidity Index [38] and binary flags for congestive heart failure, chronic kidney disease, chronic obstructive pulmonary disease, and diabetes from discharge diagnosis codes in `diagnoses_icd`. These features are static per stay and provide baseline severity context.

### 3.5.6 Respiratory Support, Ventilator Settings, and GCS (58 Columns)

Respiratory support features (34 columns) capture fraction of inspired oxygen (FiO2), supplemental oxygen flow rate, and ventilation status (invasive, non-invasive, and any-active flags). Each continuous signal receives the standard numeric family, and the ventilation binary signals receive the activity-pattern family. An SpO2 to FiO2 (SF) ratio is derived where both values are available, providing a non-invasive proxy for oxygenation efficiency. All SF-derived features are included.

Ventilator settings features (15 columns) capture positive end-expiratory pressure (PEEP), tidal volume, and minute ventilation from `chartevents`. Each receives a reduced feature family including current-hour value, last known value, six-hour rolling mean, observation flag, and hours-since. These settings reflect respiratory failure severity and can interact with cardiovascular function through changes in intrathoraic pressure affecting venous return and cardiac output.

Glasgow Coma Scale features (9 columns) capture the GCS total score, a standard neurological assessment which ranges from 3 (indicating deep coma) to 15 (fully alert) [39]. The GCS total receives the numeric family including current-hour value, observation count, observation flag, baseline, last value, six-hour rolling mean, delta, percentage delta, and hours-since. GCS serves as a severity proxy and may reflect the clinical deterioration that occurs with hemodynamic instability.

### 3.5.7 Demographics (4 columns)

Four static demographic features are included for every stay: age at ICU admission, sex, admission type, and ICU unit type. These features are constant across all hours within a stay and are always included in every model variant regardless of which signal blocks have been selected. Categorical features (sex, admission type, ICU unit) are encoded as integer codes for tree-based models, as they handle categorical splits natively.

### 3.5.8 Handling Missingness: Observation-Pattern Features as Signal

The feature engineering strategy described above does not impute missing vital signs. Instead, missingness is preserved in the feature matrix and represented through the observation-pattern features embedded in every feature family. For each signal, the binary observation flag distinguishes the measured hours from the unmeasured, the hour-since counter then indicates the recency of the last observation, and the within-window counts indicate the measurement density. These features help the model to learn associations between the measurement patterns and outcome risk, consistent with the observation framework as discussed in 2.3.4. [29]

The tree-based models used in this study (HistGradientBoostingClassifier and XGBoost) handle missing values natively by learning the optimal split directions for missing data during training [40, 41]. This means that the model can route a patient-hour with missing MAP along a different decision path than one with observed MAP without the need of imputation. The complementary mechanisms of native NaN handling and explicit observation pattern features help the model get information content of missingness.

The only exception to the no-imputation rule are the laboratory values, where the last-known-value feature uses causal forward-fill from prior hours. This is justified as laboratory results are not sampled as frequently (often every 6 to 12 hours) and the most recent result typically remains the best available estimate of the current state for clinical decision-making. Importantly, this forward-fill is applied only to the last feature variant and not the 1 hour value variant, which remains missing if no specimen was collected in the current hour. This forward fill also does not cross the onset bounday, ensuring it cannot create leakage.

## 3.6 Feature Simplification

The full feature set contains 353 columns, which produces a large training matrix when multiplied by 3.6 million valid rows. In the environment with 24GB, loading and training on the full feature matrix nears or exceeds the available capacity, particularly for models that require the entire dataset to reside in memory at once. To manage this, an optional feature simplification step is available that reduces the number of columns by removing feature variants that are less likely to carry unique information.

The simplification works by keeping only a core subset of the engineered variants for each signal. Specifically, it retains the current-hour value, the most recent prior value (last), the observation flag, the current-hour observation count, the six-hour rolling mean, any binary activity flags (such as whether a vasopressor was active), the six-hour any-active flag, hours-since-last observation counters, six-hour rolling sums, and the raw values of the core vital signs (MAP, SBP, DBP, HR, SpO2, RR, Temperature). It drops the patient baseline, the change-from-baseline features (delta and percentage delta), and the more detailed rolling statistics (minimum, maximum and standard deviation over six hours). The rationale is that the rolling mean already captures the central tendency of recent values, and the current-hour value combined with the last-known value implicitly encodes trend information, so the removed variants are partially redundant with what remains,

This simplification is applied by default in the ablation runs to ensure that all variants can be trained in the memory budget. It can be disabled using a command-line flag when running on hardware with sufficient memory. The compressed ablation suite runs without simplification, retaining the full 353 column feature set. Comparing results across the two suites therefore also serves as an indirect test of whether the dropped features carry meaningful additional information.

## 3.7 Data Splitting

The dataset is split into training, validation and test sets with a ratio of 70/15/15 percent respectively. Splitting is performed at the stay level, meaning that all hours from a single patient's ICU stay are assigned to exactly one partition, so the model is always tested on patients it has never seen during training. The split uses a grouped shuffle split. This split is performed using a fixed random seed and the resulting stay-identifier assignments are saved as a JSON artifact that is reused across all experiments without modification. This ensures that every model variant, ablation configuration, and evaluation run operates on identical data partitions.

If hours from the same ICU stay appeared in both training and test sets, the model could exploit patient-specific patterns learned during training to achieve artificially high test performance. This form of data leakage has been identified as a common methodological failure in clinical machine learning studies [42]. By assigning all hours from a given stay to exactly one partition, the split guarantees that test performance reflects generalisation to unseen patients rather than memorisation of within-patient trajectories.

The validation set helps in both threshold calibration and early stopping. The alarm threshold for each model is selected on the validation set to achieve the target 10 percent alarm rate, and this threshold is frozen before test-set evaluation. Tree-based models use the validation set for early stopping to prevent overfitting. No hyperparameter selection or model architecture decisions are made using the test set.

## 3.8 Models

Three model architectures were used in this study, spanning a linear baseline and two gradient boosted tree ensembles. All models are trained on the same stay-level split, use the same feature matrix, and are evaluated under the same fixed alarm budget protocol. The choice of model architectures was motivated by two requirements. First, the ability to handle missing values without imputation (forward fill of values), and the ability to scale to the full 3.6 million row training set within memory constraints of the environment.

### 3.8.1 Logistic Regression Baseline Methods

The initial baseline is a logistic regression model. Logistic regression is one of the simplest classification methods in machine learning. It estimates the probability of a positive outcome (in this case, that sustained hypotension will begin within the next three hours) by computing a weighted sum of all input features and then passing that result through a mathematical function called a sigmoid, which squashes the output into a value between 0 and 1 to be interpreted as a probability. Each feature receives a single weight, and the model learns these weights during training so that higher-risk hours receive higher scores. Since the relationship between features and outcome is captured by a single weighted sum with no interactions, logistic regression is a linear model.

Two additional settings control how the model trains. The first is L1 regularisation, which is a penalty that is applied during training that encourages the model to set the weights of irrelevant or redundant features to exactly zero. This performs automatic feature selection, features that do not contribute useful information are ignored rather than adding noise to the predictions. The strength of this penalty is controlled by a parameter **C** (set to 0.1); smaller values means stronger regularisation and therefore more aggressive feature removal. The second setting is the choice of optimisation algorithm used to find the best weights. This model uses SAGA Solver, which is an algorithm for handling large datasets with L1 regularisation. Both of these are configured natively in the scikit-learn library [43]. The class weight parameter is set to "balanced", which automatically adjusts the training objective to pay more attention to rare classes (which are the hours before a hypotension event) in proportion to how infrequent it is, so that the model does not learn to predict "no event" for every hour. The maximum number of training iterations used were 2000 with convergence tolerance of 1e-3.

Since logistic regression does not natively handle missing values (it requires a value for every feature in every row), it is the one model in this study that requires imputation. A preprocessing pipeline replaces missing numeric features with the median value from the training set, a strategy called median imputation. This is a departure from the general principle described in Section 3.5.8, where missingness is preserved as informative signal. The compromise is accepted here because logistic regression serves only as a baseline; the primary models (HGB and XGBoost) handle missing values natively and do not require any imputation. Categorical features (sex, admission type, ICU unit) a process called one-hot encoding, with a value of 1 if present and 0 otherwise. Unknown categories encountered in validation or test data are ignored rather than causing an error. Both transformations are fitted on the training set only and applied without refitting to the validation and test sets, so that no information from the test data leaks into the preprocessing. This pipeline is wrapped in a scikit-learn pipeline object to do this separation automatically.

This logistic regression serves as a sanity check rather than a competitive model. Its purpose is to confirm that a predictive signal exits in the feature set and that the evaluation pipeline does indeed produce plausible results. A linear model cannot easily capture the complex relationships present in ICU data, for example how a patient's current blood pressure interacts with its recent trend, how long it has been since the least measurement and if vasopressors are being administered simultaneously. These limitations thus motivated the move to tree-based models, which can learn such feature interactions natively.

### 3.8.2 Histogram Gradient Boosting (HGB)

The primary tree-based model used in this study is scikit-learn's HistGradientBoostingClassifier [43], which implements a variant of gradient boosting called Histogram-based gradient boosting. This approach follows the algorithmic principles introduced by Ke et al. [41] and serves as the methodological ancestor of scikit-learn's implementation. The core idea behind the "histogram" part is speed optimisation: instead of examining every individual feature value to find the best point at which the patients should be split into two groups, the algorithm first sorts each feature's values into a fixed number of bins (like a histogram) and then searches only across bin boundaries. This binning step substantially reduces computation time while maintaining accuracy, which makes it feasible to train on millions of rows.

A key advantage of this model for the present study is its native handling of missing values. When the algorithm encounters a missing entry during training, it does not require that value to be filled in first. Instead, it evaluates both possible directions at the split point (sending the patient-hour with the missing value to the left branch or the right branch of the decision tree) and chooses whichever direction produces a better prediction. This means that patient-hours where MAP or laboratory values were not observed are handled directly by the model without requiring any imputation or filling-in step. The model thus learns distinct behaviour for hours when a measurement is present versus hours when it is absent, which particularly relevant for the MAP-missing analysis that is central to this thesis.

The hyperparameters, which control the configuration of how a model trains, are set as follows.

- Learning rate is 0.03, which determines how much influence each new tree has on the overall prediction. Smaller values mean the model takes smaller learning steps, which generally leads to better final performance but requires more trees.
- The maximum number of iterations (trees) is 12000.
- The maximum tree depth is 3, meaning each individual tree can make at most three sequential decisions, which keeps each tree simple and prevents the model from memorising individual training examples.
- The maximum number of leaf nodes is 63, and the minimum number of patient hours required in each leaf is 50 (the final prediction group at the bottom of a tree). This minimum ensures that the model does not create prediction groups based on too few examples, which may make them unreliable.
- The L2 regularisation strength is 1.0, which is the penalty that discourages the model from making extreme predictions based on small amounts of evidence.

Early stopping is enabled to prevent the model from training for longer than necessary. The algorithm automatically sets aside 10 percent of the training data as an internal monitoring set. After each new tree is added, the model then checks whether the predictions on the aforementioned monitoring set have improved. If there is no improvements for 200 consecutive trees, then the training stops automatically as a safeguard against overfitting (wherein the model would learn patterns specific to the training data). The loss function used is log-loss, also known as binary cross-entropy, which is the standard objective for models that output probability estimates for binary (yes/no) classification tasks.

Categorical features (sex, admission type, ICU unit) are encoded as integer category codes and passed directly to the classifier. Unlike logistic regression, no one-hot encoding is required because the histogram-based algorithm handles integer-valued categorical features efficiently through its binning mechanism, treating each integer code as a separate bin.

### 3.8.3 XGBoost

The second tree-based model used in this study is XGBoost [40], which stands for Extreme Gradient Boosting. It is one of the most widely used gradient boosting libraries in both research and industry. Like HGB, XGBoost builds an ensemble of decision trees sequentially and handles missing values natively by learning default split directions for absent entries. XGBoost additionally includes a sparsity-aware algorithm that is specifically designed to be efficient when the input data contains many missing values, which is common in clinical datasets where different measurements are taken at different times. A practical advantage of XGBoost is that it can run on a GPU for faster computations and can train substantially faster than a standard HGB model which trains on CPU.

The hyperparameter for full training are as follows:

- The maximum number of trees is 700
- The learning rate is 0.05, slightly higher than HGB's 0.03, which means each tree has slightly more influence on the final prediction.
- The maximum tree depth is 3, matching HGB
- The subsample ratio is 0.8, meaning each tree is trained on a random 80 percent sample of the training data rather than all of it. This randomness reduces overfitting by ensuring that each tree sees a slightly different version of the data. Similarly, the column subsample ratio is 0.8, meaning each tree considers only 80 percent of the available features, which further promotes diversity among trees and prevents any single feature from dominating the entire ensemble.
- The minimum child weight is 5.0, which controls how many patient-hours must be affected by a potential split before the algorithm considers it worthwhile; higher values make the model more conservative.
- The L2 regularisation strength is 1.0.

For XGBoost, the training objective is binary logistic loss, and two evaluation metrics are computed on the validation set during training. AUROC (Area Under the Receiver Operating Characteristic Curve), which measures how well the model ranks high-risk hours above low risk hours on a scale of 0.5 (random guessing) to 1.0 (perfect ranking). AUPRC (Area Under the Precision-Recall Curve), which is a stricter measure that is more informative than AUROC when the positive class is very rare since it focuses on how precise the model's high-confidence predictions are. Early stopping is enabled with a patience of 50 rounds (if neither evaluation metric improves for 50 consecutive trees, then training stops). The tree construction method is histogram-based, and the device is set to CUDA when a compatible GPU is detected, falling back to CPU otherwise.

For backend auto-selection (described in Section 3.8.8), a lightweight benchmark comparison is run before full training is done to determine whether HGB or XGBoost performs better on the current feature set. This benchmark trains a reduced version of each model (180 trees, learning rate 0.08) on a subsample of the validation data and compares their recall at the target alarm rate. Recall, in this context, is the proportion of the truly positive hours that the model successfully identifies as high risk. Whichever backend achieves higher Recall is selected, with AUROC as a tiebreaker if Recall is identical. The selected backend is then used for full training on the complete training set.

### 3.8.4 Class Imbalance Handling

The dataset has a positive class prevalence of 0.54%. This means that for every single hour where hypotension is about to begin, there are approximately 185 hours where it is not. This extreme rarity of positive cases is known as class imbalace, and it is a common challenge in clinical prediction tasks. Prior to adjustment, gradient boosted models usually tend to learn that the safest strategy is to predict "no event" for almost every hour, as this would be correct 99.5% of the time. The resulting probability estimates cluster near zero for almost all patient hours, making it difficult to find a meaningful threshold that separates genuinely high-risk hours from the rest.

To counteract this, both tree-based models are configured to give more importance to rare positive hours during training. For HGB models, this is done through instance-level sample weights. Each positive (hypotension imminent) hour is assigned a weight equal to the ratio of negative to positive training examples,which is approximately 185. Each negative hour retains a weight of 1. The effect is that when the model makes an error on a positive hour, the penalty in training is 185 times larger than if the error had occurred on a negative hour. This rebalancing ensures that the model does not ignore the rare but clinically important positive cases. These weights are passed to the model's fit method via the `sample_weight` parameter.

For XGBoost, the equivalent mechanism is the `scale_pos_weight` parameter, which is set to the same ratio of approximately 185. Rather than assigning individual weights to each training example, this parameter globally scales the error signal (called the gradient) from all positive instances during tree construction. The mathematical effect is the same, as positive hours contribute proportionally more to the model's learning progress.

For the logistic regression baseline, class imbalance is handled through the `class_weight="balanced"` setting, which automatically reweights the loss function in proportion to the inverse frequency of each class. The effect is equivalent to the sample weighting used in the tree models: errors on rare positive hours are penalised more heavily during training. All three model types therefore receive some form of class imbalance correction, though the mechanism differs (explicit sample weights for HGB, a built-in scaling parameter for XGBoost, and automatic inverse-frequency reweighting for logistic regression).

### 3.8.5 Training Procedure

The training procedure for each ablation variant goes through five steps. First, the feature matrix is loaded from the dataset file in manageable chunks rather than all at once, to avoid bloating the environment memory (set to 24GB). During loading, only valid rows are retained, and all numeric values are converted to 32-bit floating point precision, which use half the memory compared to the default 64-bit representation. Categorical columns (sex, admission type, ICU unit) are encoded as integer representations. Before proceeding, the pipeline estimates the total memory that the feature matrix may require, if this estimate exceeds 8GB or 70 percent of available system memory (configured to our test environment), the training is aborted with a warning.

Second, the selected model backend (HGB or XGboost) is instantiated with the hyperparameters described above.

Third, the model is fitted on the training partition. The validation set is provided as a monitoring set for early stopping: after each tree is added, the fitted model is evaluated on this held-out data, and training stops if no improvement occurs within the patience window.

Fourth, the alarm threshold is calibrated. The trained model produces a continuous risk score for each validation-set hour. To convert these scores into binary alarm decisions, a threshold is chosen. This threshold is set to achieve the target 10 percent alarm rate on the validation set, which means selecting the score value at the 90th percentile of the validation score distribution: the point above which only 10% percent of validation set hours fall. This threshold is then frozen and applied to the test set, ensuring that the test evaluation reflects realistic deployment performances.

Fifth, the trained model and its associated threshold are saved to disk using joblib serialisation, written first to a temporary file and then moved into place in a single operation so that a crash mid-save cannot produce a corrupted artifact. Saving the model artifacts enables all downstream analyses to be performed without retraining.

### 3.8.6 Ablation Study Design

With 20 feature blocks and 353 columns available, a natural question is which of these signals actually help the model predict hypotension and which are redundant or even harmful. Rather than training a single model on everything, this thesis uses an ablation study: a systematic experiment in which feature groups are added or removed one at a time so that the effect of each group on performance can be measured in isolation. The feature families described in Table3.1 define the units of addition and removal.

### 3.8.7 Block-Based Feature Group Registry

All 353 features are organised into 20 named blocks, where each block corresponds to a clinically coherent group of signals. For example, the bp block contains all 34 blood pressure features (MAP, SBP, DBP with their rolling statistics, trends, and observation patterns), the labs block contains 117 laboratory features, and the sedation block contains the 14 sedation and analgesia features. These blocks are defined in a central registry so that any experiment can specify which blocks to include simply by name.

Blocks can also be combined into groups using macros, the vitals macro can expand to include heart rate, SpO2, respiratory rate, and temperature blocks together. The `all_non_bp` macro expands to include every block except blood pressure (vitals, labs, urine, vasopressors, vasopressor dose, fluids, balance, respiratory support, ventilator settings, GCS, sedation, interventions, infection proxy, acid-base, CBC inflammation, comorbidity, and demographics. These macros make it straightforward to define variants without having to list every individual block.

### 3.8.8 Two Ablation Suites

Two complementary suites of ablation experiments were run, each with 15 variants.

The first suite (auto-suite) is designed for broad coverage with minimal manual intervention. It uses feature simplification described in 3.6 to fit within memory, and then automatically selects between HGB or XGBoost for each variant by running the benchmark comparison described in section 3.8.3. The auto-suite includes the single-block isolation variants (for example, vitals_only trains a model using only the four non-BP vital signs, and labs_only trains using only laboratory features) as well as key composite variants (see more detail in section 3.8.9). Its purpose is to answer two questions,

1. how much predictive information does each block contain on its own? and
2. what is the best achievable performance when blood pressure features are excluded entirely?

The second suite is called compressed15 (fifteen variants using the full, uncompressed feature set, the name refers to compressing the experimental design into 15 carefully chosen combinations rather than compressing the features). It uses HGB exclusively with the full (unsimplified) feature set. It is designed around an additive composition logic; starting from blood pressure only, features are progressively added to measure whether each addition improves or degrades performance. Its purpose is to test whether richer feature sets improve fallback performance when MAP is missing, and also checks whether adding more features on top of blood pressure helps or hurts the model.

Both suites use the same locked evaluation protocol: the same data split, the same 10 percent alarm rate target, and the same three-regime threshold calibration (global, MAP-missing, MAP-observed) to ensure all 30 variants are directly comparable.

### 3.8.9 Variant Composition

The full list of variants across both suites spans single-block isolates (such as pressors_only, resp_support_only, acid_base_only, and cbc_inflammation_only), composites that combine blood pressure with one additional block (such as bp_plus_vitals, bp_plus_labs, bp_plus_gcs), richer non-BP composites (such as vitals_labs_only, vitals_labs_pressor_fluids_gcs_resp), and the two extremes of bp_only (blood pressure features alone, a strong and practically attractive single-model configuration) and all_non_bp (every feature except blood pressure, the best available fallback). The bp_only variant is trained twice, once with each backend, and the better-performing version is selected as primary model.

## 3.9 Evaluation Framework

In a traditional classification study, models are usually compared using threshold-free metrics that summarise how well the model ranks positive examples above negative ones across all possible decision boundaries. While these ranking metrics are useful for comparing discrimination, they do not answer the question that matters most for an alarm system: how many alarms will the system actually generate, and how many events will it miss at that workload? In an ICU environment, every alarm competes for a clinician's limited time and attention, and alarm overload has been formally recognised as a patient safety problem [4, 5].

This thesis therefore evaluates models not only as classifiers but as components of an alarm policy. The distinction is between model performance, which is determined by how well the model ranks risk across all possible thresholds, and operational performance, which is determined by how well the model behaves at specific thresholds that have been chosen to produce a number of alarms. The evaluation framework described in this section specifies how thresholds are chosen, what metrics are reported at those thresholds, and how performance is stratified by whether blood pressure data is available at the time of prediction. This framework, combined with alarm management principles described in the Background chapter 2 [10], ensures that model comparisons reflect meaningful differences rather than abstract ranking improvements that may not translate into better alarm behaviour.

### 3.9.1 Fixed Alarm Budget Thresholding Protocol

Each model in this study produces a continous risk score between 0 and 1 for every eligible patient-hour. To convert these scores into binary alarm decisions, a threshold must be chosen, any hour with a risk score above the threshold triggers an alarm, and any hour below it does not. The choice of this threshold determines the tradeoff between sensitivity (how many true events are caught) and alarm burden (how many total alarms are generated).

Rather than selecting a threshold that maximises some composite metric, this study uses a fixed alarm budget approach. The target alarm rate is set to 10% of patient hours, which means that the system is designed to alarm on about one in every ten hours. This rate is chosen as an operating point that balances detection against workload, which is set high enough to catch a meaningful amount of events but low enough as to not overwhelm clinical staff with alerts.

The threshold is determined as described. After the model is trained, it produces risk scores for every hour in the validation set. The threshold is then set to the value at the 90th percentile of this score distribution, which is the point above 10% of validation hours fall. This threshold is then frozen and applied without any modification to the test set. The model's test performance is evaluated at only this pre-determined threshold, without any adjustment. This protocol ensures that the test evaluation reflects what would happen if the system was deployed with the threshold that was calibrated before seeing any of the test data, which is a more realistic simulation of the deployment setting rather than optimizing the threshold on the test set [13].

Each model variant receives its own threshold, which are calibrated independently on the validation set. This means that the models with different score distributions are each evaluated at the same effective alarm burden (10 percent), making their performance directly comparable. Without this standardisation, a model that produces higher scores overall would seem to have better recall simply due to its scores exceeding an arbitrary fixed threshold rather than it genuinely ranking risk.

### 3.9.2 Hourly and Event-Level Metrics

Performance is reported at two levels of granularity: hourly and event level.

At the hourly level, two ranking metrics are reported (threshold free). The first is AUROC (Area Under the Receiver Operating Characteristic curve), which measures how well the model is in separating the truly positive hours from the truly negative hours across all possible thresholds. An AUROC of 1.0 means the model perfectly ranks every positive hour above every negative hour, while 0.5 means it is as good as random guessing. AUROC is useful for comparing overall discrimination between models, but it can paint an optimistic picture when the positive class frequency is rare, because even a model that produces many false alarms can achieve a high AUROC if it ranks the few positives near the top [11]. The second ranking metric is AUPRC (Area Under the Precision-Recall Curve), which focuses specifically on the model's behaviour among the hours it scores the highest. Precision measures what fraction of alarmed hours are truly positive. AUPRC is more sensitive than AUROC in highly imbalanced settings, with a prevalence of 0.54 percent, a random classifier would achieve an AUPRC of approximately 0.005, so any value substantially above this threshold indicates a real predictive signal [12].

At the fixed 10% alarm rate, two threshold-specific hourly metrics are reported. Recall at the alarm rate (also called sensitivity) measures the proportion of truly positive hours that receive an alarm. Precision at the alarm rate (also called positive predictive value) measures the proportion of alarmed hours that are truly positive.

In addition to hourly metrics, event-level metrics are reported. An event is defined as a single hypotension onset. An event is considered as detected if the model produces at least one alarm during the prediction horizon window (upto 3 hours) before the onset hour. Event recall is the proportion of events that are detected, and event miss rate is the complement (the proportion of events that receive no alarm at all before onset). Event-level metrics are more clinically meaningful than hourly metrics because clinicians need at least one warning before a deterioration episode, not an alarm at every single hour. A model that misses some individual positive hours but still triggers at least once before every event may be more useful than one that catches every positive hour for some events that misses others entirely.

### 3.9.3 Lead Time and Cooldown-Based Alert Burden

Two additional operational metrics are also reported to characterise how the alarm system would work in practice.

Lead time measures how early the first alarm fires before the onset of a sustained hypotension event. For each detected event, the lead time is computed as the number of hours between the first alarm within the prediction horizon window (of 3 hours) and the onset hour itself. Since the prediction horizon is of 3 hours, the maximum possible lead time is three hours and the minimum is one hour. Lead time is reported as a distribution across all detected events, with the median, mean and interquartile range. A system that detects most events but only raises alarms one hour before onset provides less clinical utility than one that raises alarms two or three hours ahead, because earlier warnings give clinicians more time to reassess and intervene.

Cooldown-based alert burden measures the effective number of distinct alerts a clinician would receive per day, after accounting for the fact that consecutive alarms for the same patient within a short window would typically be suppressed. A cooldown period is the time interval after an alarm fires during which subsequent alarms for the same patient are suppressed. This converts the raw hourly alarm rate into a more realistic estimate of actual workload. Alert burden is reported at cooldown values of 0 (no suppression), 3, 6, and 12 hours, expressed as the average number of distinct alerts per 24 equivalent hours of monitoring. This metric bridges the gap between the abstract alarm rate (a percentage of hours) and the question a clinician or manager would ask: how many times per day will this system interrupt me?.

### 3.9.4 MAP-Missing Evaluation

An important evaluation design choice in this thesis is the stratification of all performance metrics by whether mean arterial pressure was observed at the time of prediction. This stratification is motivated by a deployment concern that is central to the research question which is that MAP is absent in approximately 56% of patient-hours in the dataset, and as the results show, this absence has a huge impact on how alarm thresholds behave.

The problem arises because the model's risk scores have different distribution depending on whether MAP is available. When MAP is recorded, the blood pressure features carry a strong predictive signal and where the model produces scores, high scores for high-risk hours and low scores for stable hours. When MAP is absent, the model must rely on weaker indirect signals (vital sign trends, observation patterns, laboratory values), and the resulting scores tend to cluster in a lower, narrower range. A threshold that is calibrated on the full validation set (which contains both MAP-observed and MAP-missing hours mixed together) will be dominated by the MAP-observed majority hours, producing a cutoff that is too high for the MAP-missing subset. The result is that patient-hours without recent MAP observations may receive essentially zero alarms, despite the patient's risk. This is a **coverage disparity**, a substantial portion of patients are effectively invisible to the alarm system not because they are low risk, but because their monitoring pattern produces lower scores that fall below a threshold calibrated on a population with a different score distribution.

To address this, three separate thresholds are calibrated for each model. The first is the global threshold, calibrated on the full validation set to achieve the target 10% alarm rate across all hours. This is the standard evaluation and serves as a baseline. The second is the MAP-missing threshold, calibrated only on the subset of validation hours where MAP was not observed, again targeting a 10% alarm rate within that subset. The third is the MAP-observed threshold, calibrated only on the subset of validation hours where MAP was observed, targeting 10% within that subset. Each threshold is frozen after calibration and applied to the corresponding test subset without modification.

The MAP-missing mask is determined using a priority-ordered lookup which means that if a column explicitly marking MAP being available exists then it is used directly, otherwise the binary MAP observed feaure is used. As a final fallback, hours where raw MAP value is absent are classified as MAP-missing.

For each model, performance is reported across a matrix of subsets and thresholds. The full cohort is evaluated under the global threshold. The MAP-missing subset is evaluated under both the global and MAP-missing calibrated thresholds. This produces a reporting matrix that isolates two distinct effects, which is the calibration effect (whether the model's ranking ability differs between subsets). A warning is applied when the number of positive hours in a validation subset falls below 25, as threshold calibration becomes unstable with very few positives.

### 3.9.5 Evaluation Summary

The evaluation framework described in this section combines three different layers of assessing models. The first is threshold-free ranking (AUROC, AUPRC), which measures the overall discrimination reagrdless of alarm rate. The second layer is the threshold-specific operational metrics (recall, precision, event recall, event miss rate, lead time, cooldown alert burden) at the fixed 10% alarm budget, which measures how the system would behave in practice at a realistic workload. The third layer is the MAP-missing stratified evaluation, which tests whether the system provides equitable covrage across patients with different monitoring patterns.

Together, these three layers ensure that the results presented in the next chapter reflect not only how well a model ranks risk but how it could function as an alarm system in an ICU where blood pressure is sometimes unavailable. These results will be discussed in the next chapter.

# 4 Experiments and Results

This chapter presents all experimental results. It begins with the cohort and missingness statistics, then reports baseline blood pressure model performance, expanded feature ablation results, the MAP-missing coverage disparity analysis, gated policy comparisons, and error analysis with patient case studies. All models follow the locked protocol described in Methods: the same stay-level split, the same 10 percent alarm budget, and the same three-regime threshold calibration.

## 4.1 Experimental Setup

This chapter presents the results of the experiments conducted in this study. Here is a recap of the key elements of the locked protocol every experiment has.

All models are trained, validated and tested on the same stay-level data split described in Section 3.7. The prediction horizon is of three hours, at each eligible hour, the model must estimate whether sustained hypotension will begin within the next one to three hours. Hours before the sixth hour of ICU admission are excluded from prediction because they serve as the baseline window for computing patient-specific reference values. All hours at or after hypotension onset are excluded entirely, so the models are never asked to detect an event that has already started.

Every model is evaluated under the fixed 10% alarm budget described in Section 3.9.1 . For each variant, a threshold is calibrated on the validation set to produce alarms on 10% of hours, and that threshold is frozen before any test-set evaluation. Performance is then reported at this operating point, not at an optimized or chosen threshold.

Two ablation suites are run. The auto-suite trains 15 model variants using automatic backend selection between XGBoost and HGB, with the simplified feature set to stay within the environment memory budget (24GB). The `compressed15` suite trains 15 variants using HGB exclusively with the full 353-column feature set. Both suites use the same three calibration regimes: a global threshold calibrated on all validation hours, a MAP-missing threshold calibrated only on validation hours without MAP observations, and a

MAP-observed threshold calibrated only on hours with MAP observations. This stratified evaluation is the mechanism through which the core thesis is demonstrated (the MAP-missing coverage disparity).

## 4.2 Cohort, Label and Missingness Statistics

The dataset used across all experiments contains 94,259 unique ICU stays taken from MIMIC-IV v3.1, covering the first 48 hours of each stay on an hourly grid. This results in 4,524,432 (4.5 Million) total patient-hour rows. After applying the valid mask (which removed post-onset hours and hours before the baseline window closes at hour six), 3,590,802 (3.5 Million) rows remain eligible for prediction. Of these, 19,520 carry a positive early warning label, meaning that sustained hypotension onset occurs within three hours of that hour. This gives a valid-row prevalence of 0.54 percent, or roughly one positive hour for every 184 negative hours.

Table 4.1: Dataset summary. The cohort comprises all adult ICU stays in MIMIC-IV v3.1, restricted to the first 48 hours per stay. Valid rows exclude post-onset hours and hours before the baseline window closes at hour 6.

| **Property** | **Value** |
|---|---:|
| Data source | MIMIC-IV v3.1 |
| Observation window | Hours 0–47 (first 48h) |
| Unique ICU stays | 94,259 |
| Total patient-hour rows | 4,524,432 |
| Valid rows (eligible for prediction) | 3,590,802 |
| Positive early warning labels | 19,520 |
| Valid-row prevalence | 0.54% |
| Prediction horizon | 3 hours |
| Baseline window (excluded from prediction) | Hours 0–5 |
| Feature columns | 353 |
| Feature blocks | 20 |

The data is split at the stay level into training, validation and test sets in ratio of 70/15/15. Since the split is performed by stay rather than by individual hours, each patient's complete ICU trajectory appears in exactly one set. The training set contains 65,981 stays, the validation set 14,139 and the test set 14,139 (Table 4.2). Positive label counts and prevalence are consistent across the sets, confirming that the random split did not introduce any meaningful imbalance between sets.

The most consequential data characteristic for this study is the rate at which the Mean Arterial Pressure values are missing. Across all valid rows, MAP is absent in approximately 56% of patient-hours (55.7% in the test set) . This means that more than half the time, the model is asked to make a prediction without its single most informative input feature. This

Table 4.2: Stay-level data split. All hours from a given ICU stay appear in exactly one partition. Proportions are 70/15/15, assigned using a fixed random seed. Positive counts and prevalence are consistent across partitions.

| Partition | Stays | Total Rows | Valid Rows | Positive Rows | Prevalence (%) |
|---|---|---|---|---|---|
| Train (70%) | 65,981 | 3,167,088 | 2,514,402 | 13,664 | 0.54 |
| Validation (15%) | 14,139 | 678,672 | 538,347 | 2,928 | 0.54 |
| Test (15%) | 14,139 | 678,672 | 538,053 | 2,928 | 0.54 |
| Total | 94,259 | 4,524,432 | 3,590,802 | 19,520 | 0.54 |

is not evenly distributed across patients, some stays have MAP charted nearly every hour (typically patients on arterial lines or receiving intensive monitoring), while others have MAP recorded only a few times across the full 48-hour window. This variation reflects real differences in clinical monitoring intensity rather than random data loss [29]. Table 4.3 confirms that the 56% missing rate is stable across train, validation and test partitions.

How does this affect the experiments? A model relying heavily on MAP features will output different risk scores depending on whether MAP was recently observed. If a global threshold is calibrated on the full population data (which mixes MAP-observed and MAP-missing hours together), the threshold may be set at a level that works well for MAP-observed hours but is too high for MAP-missing hours. The result, as will be shown later on, is that a large fraction of patient hours receive essentially no alarm coverage, not due to being low risk but because their monitoring pattern produces lower scores.

Table 4.3: MAP observation rates across data partitions. MAP is absent in approximately 55%-56% of all valid patient-hours, and this rate is stable across train, validation, and test sets. MAP-missing hours are defined as hours where no MAP value was charted

| Partition | Valid Rows | MAP-Observed | MAP-Missing | Missing Rate (%) |
|---|---|---|---|---|
| Train | 2,514,402 | 1,119,302 | 1,395,100 | 55.5 |
| Validation | 538,347 | 239,765 | 298,582 | 55.5 |
| Test | 538,053 | 238,338 | 299,715 | 55.7 |
| Total | 3,590,802 | 1,597,405 | 1,993,397 | 55.5 |

The feature matrix contains 353 columns spanning 20 signal blocks, as described in Section 3.5. In the auto-suite experiments, the simplified feature set reduces this to approximately 200 columns by dropping baseline, delta, and detailed rolling statistics. In the compressed15 experiments suite, all 353 columns are used.

## 4.3 Baseline Results (BP Only)

### 4.3.1 Logistic Regression Baseline

Before training tree-based models, a logistic regression was fitted using only blood pressure features to confirm that the evaluation pipeline works and that the data contains real predictive signal. As described in Section 3.8.1, this is a linear model with L1 regularisation, median imputation for missing values, and balanced class weighting.

On the test set, logistic regression achieved an AUROC of 0.880 and an AUPRC of 0.034 (Figure 4.1. For context, a random classifier on this dataset would get an AUPRC of about 0.005 (matching the 0.54 percent prevalence), so 0.034 represents a real improvement over the 0.005 random baseline, though far below the tree-based models. However, the model's limitations showed up under the fixed alarm budget. At a 10% alarm rate, recall was moderate and event-level performance was weaker than what AUROC alone would suggest. The alarm trade-off curves revealed the core tension that catching more events required a sharp increase in alarm volume, which negatively impacted precision. At 10% alarm rate, recall was 57.7% and 210 out of 1,053 events received no alarm at all, an event miss rate of 19.9%. Table (4.4) includes these numbers alongside the tree-based results.

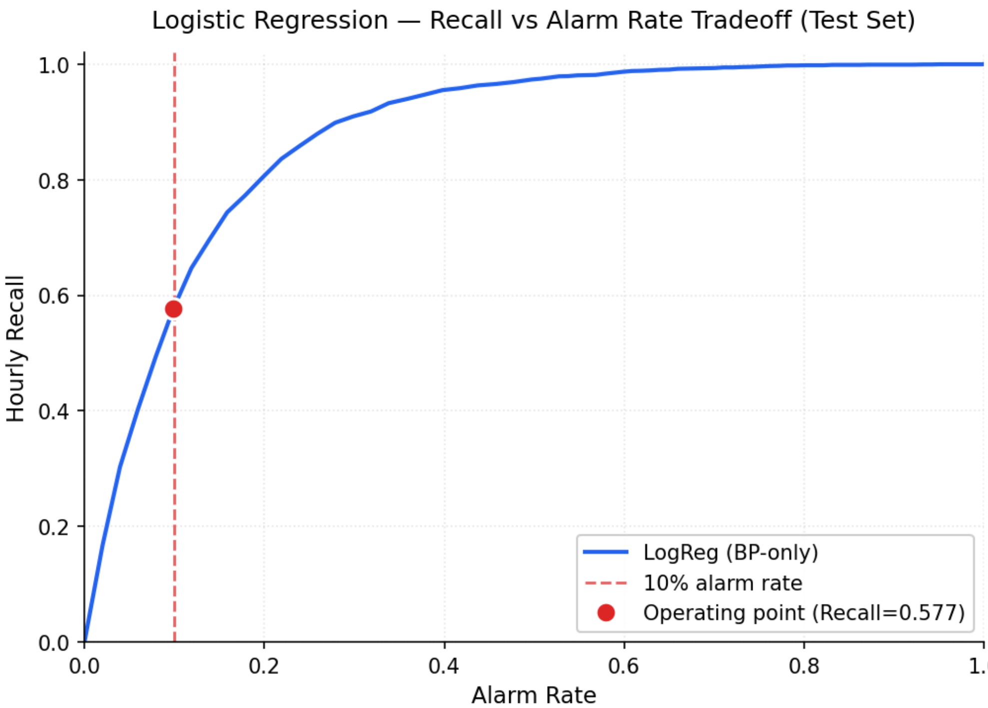


Figure 4.1: Logistic regression alarm tradeoff on the test set (BP only features). Recall increases with alarm rate but the curve is shallow compared to tree-based models. The dashed vertical line marks the 10 percent alarm budget; the red dot marks the operating point (57.7 percent recall).

The results confirmed two things. First, blood pressure dynamics are a strong signal for predicting hypotension even through simple linear combination of features. Second, the linear model cannot capture the non-linear interactions that do matter in ICU data, like how a patient's current MAP interacts with its recent trajectory, how long it has been since the last measurement, and whether the trend is accelerating. A model that treats each feature independently will miss these connections. Thus, logistic regression serves as a sanity-check model and the remaining experiments all use tree-based models.

### 4.3.2 HGB and XGBoost

The bp_only model, trained using only the 34 blood pressure features described in Section 3.5.2, is the strongest single model in this study. The two backends were compared (HGB and XGBoost) using the same feature set and the same locked evaluation protocol (Table 4.4

In the auto-suite (with the simplified feature set), both backends achieve a test AUROC of 0.945, XGBoost produced slightly higher recall at the 10% alarm rate (82.2% versus 82.1% for HGB) and was auto-selected as the primary backend based on the validation benchmark. At the event level, bp_only model caught 98.3% of all hypotension events in the test set, missing only 1.7%. Which means that out of roughly 1000 test-set events, approximately 18 received no alarm at all before onset.

In the compressed suite (which has the full 353-column feature set, HGB only), the bp_only model reached a test AUROC of 0.955, recall of 87.8% at the 10% alarm rate, and an event miss rate of 0.66% which catches over 99% of events. The difference between 0.945 and 0.955 AUROC, and between 82.2 and 87.8 percent recall, reflects both the feature simplification (the auto-suite drops baseline, delta, and detailed rolling statistics) and the backend difference (auto-selected XGBoost versus HGB). The two factors cannot be fully separated, but the dropped features likely account for most of the gap since the blood pressure block benefits from trend and deviation features. Still, even the simplified version misses fewer than 2% of events.

XGBoost's early stopping was useful across both suites. For bp_only models, training stopped at about 445 iterations out of a maximum of 700, indicating the model converged well before reaching its maximum.

### 4.3.3 Feature Importance and Lead Time

In order to understand what features the bp_only model gets its predictive signal from, permutation importance was run on the test set. Permutation importance works by taking one feature at a time and randomly rearranging its values across patient hours, breaking any

Table 4.4: bp_only model performance across backends and ablation suites, including the logistic regression baseline. The auto-suite draws from a simplified column pool of approximately 200 columns (each variant selects only its relevant blocks) and selects between XGBoost and HGB automatically. The compressed15 suite uses HGB with the full 353-column feature set. All results are on the held-out test set at the fixed 10% alarm rate.

| Backend | Suite | Features | AUROC | Recall@10% | EventRecall | EventMiss |
|---|---|---|---|---|---|---|
| LogReg | Baseline | 34 | 0.880 | 57.7% | 80.1% | 19.9% |
| XGBoost | Auto-suite | ∼200 | 0.945 | 82.2% | 98.3% | 1.7% |
| HGB | Auto-suite | ∼200 | 0.945 | 82.1% | 98.3% | 1.7% |
| HGB | Compressed15 | 353 | 0.955 | 87.8% | 99.3% | 0.7% |

real relationship between that feature and the outcome while keeping everything else intact. If this destroys performance, that feature was carrying important information.

Figure 4.2 shows the importance of the 10 most important features of the bp_only model. MAP itself is the single most important feature. After MAP, the next strongest features were MAP's rolling six-hour statistics (MAP mean, MAP minimum and MAP maximum over the preceding 6 hours). These features capture the recent trajectory of the MAP rather than its current value. SBP and DBP features followed a similar pattern but with lower importance.

Surprisingly, measurement-pattern features (`hours_since_MAP`, `MAP_observed`, and the six-hour observation count) ranked among the top 10 features. These are not physiological values but they encode how recently and how frequently blood pressure was measured. Their importance confirms that observation timing carries a predictive signal in this dataset, which is consistent with the informative missingness argument made in Section 2.3.3 [29]. So if the model relies partly on whether MAP was recently observed, then hours where MAP has not been measured will produce systematically different scores.

Static demographics features (age, sex, admission type, ICU unit) contributed very little as shuffling them barely impacted performance. This implies that the model entirely prioritizes recent physiological dynamics and monitoring patterns over patient demographics.

For the lead time, we looked at how early the first alarm alerted before each detected event. See Figure 4.3. Across the test set, the median lead time was 2.0 hours. Mean lead time was 2.2 hours with an interquartile range of [2.0, 2.0] hours. This means that for most detected events, the first alarm fires approximately two hours before onset.

We also examined how the alarm stream would translate into a real workload via cooldown suppression. Without any cooldown, the raw alarm rate is 10 percent of hours by construction, which translates to 2.4 alerts per 24 hours of monitoring . With a three hour cooldown (consecutive alarms for the same patient within three hours are grouped into a

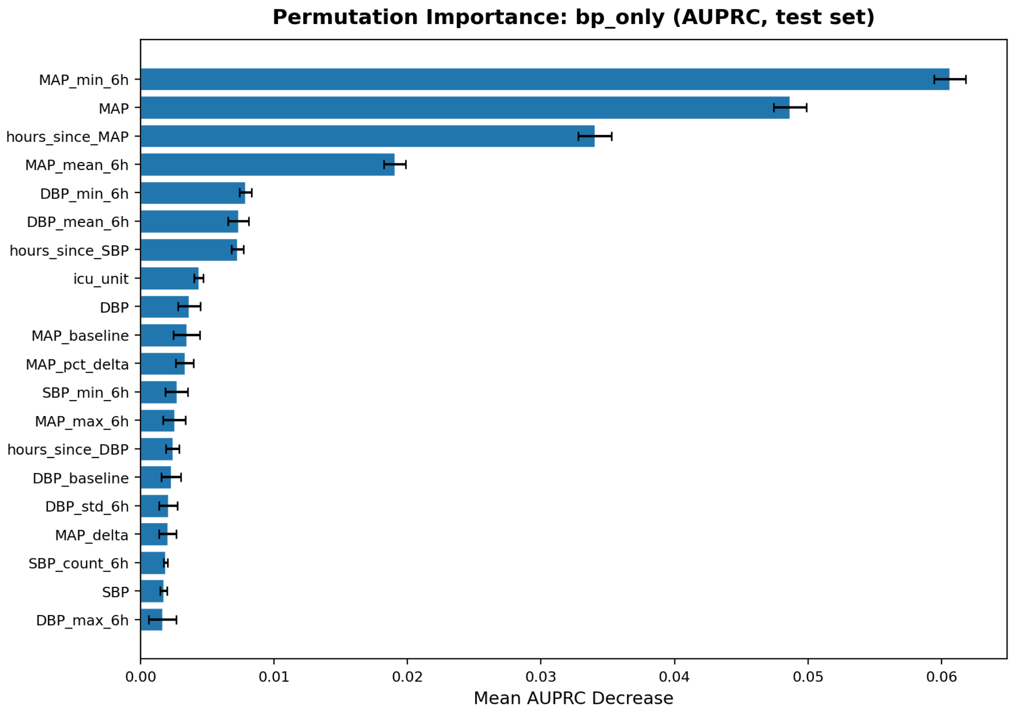


Figure 4.2: Permutation importance for the bp_only model (top 10 features), measured as decrease in AUPRC on the test set over 5 repeats.

single alert), the effective burden drops to approximately 1.1 alerts per 24 hours of monitoring. With a six-hour cooldown it drops further to about 0.72 alerts per 24 hours. These numbers suggest that the alarm system can be shaped into something that would not overwhelm clinical staff with repeated interruptions for the same patient.

## 4.4 Expanded Features Test Suite

### 4.4.1 Vitals

With the bp_only model performing so strongly, the first question is whether adding more vital signs on top of blood pressure improves anything. We tested this by training models that combine BP features with heart rate, oxygen saturation, respiratory rate, and temperature.

Adding these four vital signs to the blood pressure block produced no meaningful improvement(Table 4.5 and Table 4.6). AUROC remained at 0.954 in the compressed15 suite, recall at the 10% alarm rate barely moved, and event miss rate remained near 1.5 percent. The differences between bp_only models and bp_plus_vitals were at the noise level (fractions of a percentage point in either direction across runs). So for predicting hypotension three hours ahead, the recent blood pressure trajectory already captures what these additional vitals offer.

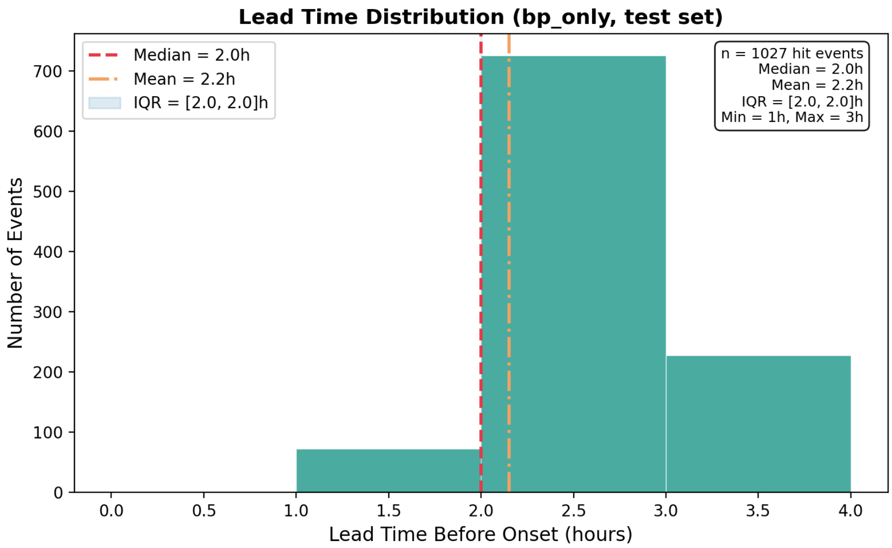


Figure 4.3: Distribution of lead times for detected hypotension events in the test set (bp_only model, compressed15 suite). Lead time is the number of hours between the first alarm and hypotension onset, bounded by the 3-hour prediction horizon

When trained in isolation (without any blood pressure features), the four vital signs performed poorly. Vitals_only achieved an AUROC of 0.733, recall of 25.7% at the 10% alarm rate, and an event miss rate of 67.3%, missing two-thirds of all hypotension events. Heart rate was the strongest contributor within this group, which makes clinical sense [3]. Tachycardia (elevated heart rate) is a common compensatory response to falling blood pressure but on its own, is not specific enough to predict sustained hypotension reliably at the hourly level.

### 4.4.2 Labs, Urine, Vasopressors

Laboratory features showed a similar pattern. Labs feature only model achieved AUROC of 0.744, slightly above the vitals_only model but still far below the blood pressure based models. At the fixed alarm rate, recall was 31.9% and event miss was 64.0%. Labs do contain a real predictive signal (an AUROC of 0.744 is well above the 0.5 random baseline) but it's a weaker, slower signal compared to direct blood pressure dynamics. Lab results arrive intermittently (6 - 12 hours in many cases), so they cannot track rapid hemodynamic changes the way hourly MAP readings can.

When labs feature were added to blood pressure features, bp_plus_labs model actually performed slightly worse than the bp_only model in the compressed15 suite. Event miss rate rose from 0.7% to approximately 4%, and AUROC dropped from 0.955 to 0.947. Adding more features did not help, but instead slightly dropped the performance. The reason being that additional lab features shifted the model's internal score distribution just enough to affect threshold calibration, without adding predictive information that BP features had not already captured.

Vasopressor features, when tested alone, showed a moderate event signal. The model trained on only Pressors features had a weak hourly discrimination but achieved roughly 28% event recall, reflecting the clinical reality that vasopressor administration often coincides with (or shortly follows) hypotension. But this is correlation with the event rather than prediction before it, and the signal does not add value on top of blood pressure features. Urine output on its own was also a weak standalone predictor, consistent with its sparse charting in the dataset.

### 4.4.3 Respiratory Support, GCS, Sedation, Derived Markers

Each of the remaining signal blocks was tested in isolation through the auto-suite. The model trained on only respiratory support features (resp_support_only) achieved an AUROC of 0.742, comparable to labs and slightly above the vitals models. Ventilator settings, GCS, sedation exposure, acid-base composites, CBC inflammation markers, and infection proxies each performed weaker individually, with AUROCs ranging from roughly 0.65 to 0.74. None of these blocks approached blood pressure performance on their own.

The value of these blocks shows up when combined into composites for the MAP-missing fallback scenario. Respiratory support indicators (FiO2, ventilation status, SF ratio) reflect escalating organ support that often precedes or accompanies hemodynamic instability. GCS changes can signal neurological deterioration that co-occurs with circulatory failure. Sedation exposure captures treatment intensity. Individually, each provides a weak and noisy signal. Together, they contributed to the incremental improvement of the all_non_bp composite over earlier, simpler non-BP baselines.

### 4.4.4 Full Non-BP Composite

The all_non_bp variant includes every feature block except blood pressure (vitals, labs, urine, vasopressors, vasopressor dose, fluids, balance, respiratory support, ventilator settings, GCS, sedation, interventions, infection proxy, acid-base, CBC inflammation, comorbidity, and demographics). Results are shown in Table 4.5. In the auto-suite (simplified features with XGBoost), it achieved an AUROC of 0.780, recall of 34.6% and an event miss rate of 58.8%.

In the compressed suite (full features with HGB), the all_non_bp model's AUROC was 0.747. This is where the information ceiling becomes clear. We tested progressively richer non-BP composites: vitals_labs_only model scored 0.746 in the compressed15 suite, vitals_labs_pressor_fluids model scored comparably, and vitals_labs_pressor_fluids_gcs_resp model reached 0.748. Adding respiratory support, GCS, sedation, and all the derived markers on top of vitals and labs bought almost nothing (AUROC moved by 0.001 to 0.002 across these variants). Across two backends and 20 feature families, the non-BP ceiling did not improve. This points to a signal limitation rather than a modeling one: without direct blood pressure measurements, the available clinical signals simply do not carry enough information to predict a MAP-defined event with the same accuracy.

For the full cohort, this means the non-BP model catches roughly 42% of events at the 10% alarm rate (which is inadequate for clinical deployment); however its role is not to replace the BP model but to serve as a fallback during MAP-missing hours, where the BP model's global threshold produces zero alarms.

### 4.4.5 Ablation Tests Summary

A complete ranking of all auto-suite variants by test-set AUROC, recall at the 10% alarm rate, and event miss rate is provided below.

Table 4.5: Auto-suite ablation results on the held-out test set, ranked by AUROC. Column counts are omitted for non-BP variants because feature simplification produces variant-specific counts depending on which blocks are selected.

| Variant | Backend | Cols | AUROC | AUPRC | Recall | EvRecall | EvMiss |
|---|---|---|---|---|---|---|---|
| bp_only (primary) | xgb | 16 | 0.945 | 0.086 | 82.2% | 98.3% | 1.7% |
| bp_only (legacy) | hgb | 16 | 0.945 | 0.086 | 82.1% | 98.3% | 1.7% |
| all_non_bp | xgb | — | 0.780 | 0.018 | 34.6% | 41.2% | 58.8% |
| vitals_labs_only | xgb | — | 0.768 | 0.018 | 32.0% | 38.6% | 61.4% |
| labs_only | xgb | — | 0.743 | 0.016 | 32.6% | 36.7% | 63.3% |
| resp_support_only | hgb | — | 0.742 | 0.014 | 28.1% | 33.1% | 67.0% |
| vitals_only | hgb | — | 0.733 | 0.014 | 25.7% | 32.7% | 67.3% |
| vent_settings_only | hgb | — | 0.687 | 0.013 | 27.9% | 31.3% | 68.7% |
| sedation_only | hgb | — | 0.674 | 0.012 | 25.2% | 27.8% | 72.2% |
| pressors_only | xgb | — | 0.665 | 0.012 | 25.6% | 27.7% | 72.3% |
| pressors_dose_only | hgb | — | 0.664 | 0.012 | 25.9% | 28.2% | 71.8% |
| infection_proxy_only | hgb | — | 0.654 | 0.010 | 22.5% | 22.2% | 77.8% |
| acid_base_only | hgb | — | 0.637 | 0.010 | 22.4% | 24.8% | 75.2% |
| cbc_inflammation_only | hgb | — | 0.617 | 0.008 | 19.3% | 19.3% | 80.7% |
| comorbidity_static_only | — | — | *skipped (static features only)* | | | | |

The compressed variants (Table 4.6) along with the cross-suite comparison for shared variants.

The key patterns are consistent across both suites. Blood pressure features dominate, non-BP blocks add minimal incremental value on the full dataset, and the non-BP composite plateaus around 0.75 AUROC regardless of adding more signal families.

Table 4.6: Compressed15 ablation results on the held-out test set (HGB backend, full feature set), ranked by AUROC. All variants are evaluated at the fixed 10% alarm rate.

| Variant | Cols | AUROC | AUPRC | Recall | EvRecall | EvMiss |
|---|---|---|---|---|---|---|
| bp_plus_gcs | 46 | 0.955 | 0.101 | 87.6% | 99.6% | 0.4% |
| bp_plus_resp_support | 71 | 0.955 | 0.099 | 88.0% | 99.5% | 0.5% |
| bp_only | 37 | 0.955 | 0.101 | 87.8% | 99.3% | 0.7% |
| bp_plus_vitals | 81 | 0.954 | 0.101 | 87.6% | 99.1% | 1.0% |
| bp_plus_uop_pressor_fluids | 71 | 0.953 | 0.100 | 87.0% | 99.0% | 1.0% |
| bp_plus_labs | 154 | 0.947 | 0.091 | 83.8% | 96.7% | 3.3% |
| bp_plus_all_new | 308 | 0.946 | 0.086 | 84.3% | 96.1% | 3.9% |
| vl_pressor_fluids_gcs_resp | 253 | 0.748 | 0.015 | 31.3% | 38.9% | 61.1% |
| all_non_bp | 319 | 0.747 | 0.015 | 30.1% | 39.2% | 60.8% |
| vitals_labs_pressor_fluids | 195 | 0.746 | 0.015 | 30.7% | 39.6% | 60.4% |
| vitals_labs_resp_support | 195 | 0.746 | 0.015 | 29.9% | 38.8% | 61.3% |
| vitals_labs_gcs | 170 | 0.746 | 0.015 | 30.5% | 39.4% | 60.6% |
| vitals_labs_only | 161 | 0.746 | 0.015 | 31.3% | 39.9% | 60.1% |
| vitals_only | 44 | 0.719 | 0.012 | 23.7% | 36.3% | 63.7% |
| labs_only | 117 | 0.705 | 0.014 | 29.9% | 35.8% | 64.2% |

Table 4.7: Cross-suite comparison for variants appearing in both ablation suites. The auto-suite uses the simplified feature set with XGBoost or HGB auto-selected; compressed15 uses the full 353-column feature set with HGB exclusively. Performance differences between suites reflect both the feature set and backend changes simultaneously, so they cannot be attributed to either factor alone.

| | Auto-Suite (simplified, auto) | | | Compressed15 (full, HGB) | | |
|---|---|---|---|---|---|---|
| Variant | AUROC | Recall | EvMiss | AUROC | Recall | EvMiss |
| bp_only | 0.945 | 82.2% | 1.7% | 0.955 | 87.8% | 0.7% |
| vitals_only | 0.733 | 25.7% | 67.3% | 0.719 | 23.7% | 63.7% |
| labs_only | 0.743 | 32.6% | 63.3% | 0.705 | 29.9% | 64.2% |
| all_non_bp | 0.780 | 34.6% | 58.8% | 0.747 | 30.1% | 60.8% |

## 4.5 MAP-Missing Robustness

This section presents the central finding of this thesis. Under standard evaluation, the bp_only model appears to perform well across the board with an AUROC of 0.955, recall of 87.8%, event miss rate under 1%. However, these aggregate numbers hide a critical deployment failure that only becomes visible when performance is stratified by whether blood pressure was actually measured at the time of prediction.

### 4.5.1 Coverage Disparity

In the test set, 299,715 out of 538,053 valid patient-hours have no MAP observation. That is 55.7% of all hours the model is asked to score. When we apply the global threshold

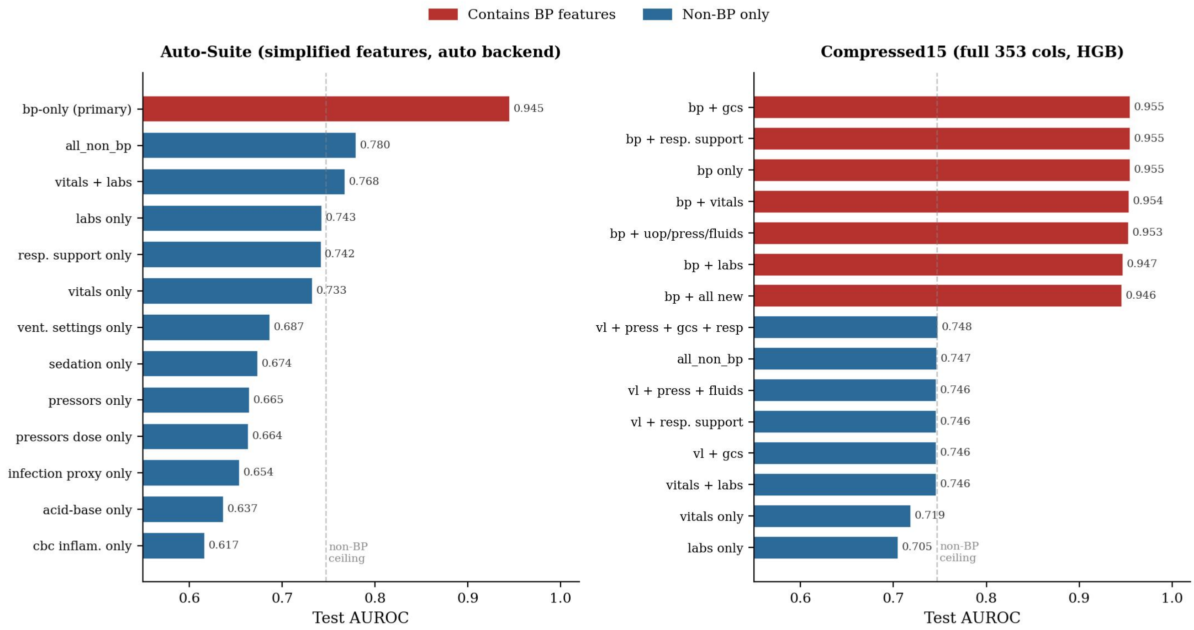


Figure 4.4: Test AUROC across all variants in both ablation suites. Red bars indicate BP containing variants. The dashed line marks the non-BP information ceiling at approximately 0.75.

(0.586, calibrated on the full validation set) to these MAP-missing hours, the alarm rate is 0.0%. Out of nearly 300,000 patient hours, the model does not fire a single alarm.

Why does this happen? It comes down to how the score distributions differ between the two subsets. When MAP is observed, the model's BP-value features (MAP, MAP_min_6h, MAP_mean_6h) carry a strong signal, and the resulting scores spread across a wide range wherein when MAP is observed, the mean score is 0.316 and the median is 0.194. When MAP is missing, scores collapse: the 90th percentile reaches only 0.018. See Figure 4.5 and Table 4.8. Only 0.006% of MAP-missing scores exceed the global threshold. In practical terms, the entire MAP-missing score distribution sits below the decision boundary.

This is not a model failure in the usual sense. The model is doing something reasonable, when its primary input is absent, it has low confidence. However the threshold, which was calibrated on a population where 44% of hours have MAP observed, sits at a level that makes sense for the MAP-observed majority and completely misses the MAP-missing minority. The 32 hypotension events that occur during MAP-missing hours in the test set receive no warning at all.

As can be seen in Figure 4.5, the MAP-observed score distribution (top panel) has a broad tail extending well past the threshold line, while the MAP-missing distribution (bottom panel) is compressed into a narrow band near zero, with the global threshold line sitting in empty space to its right.

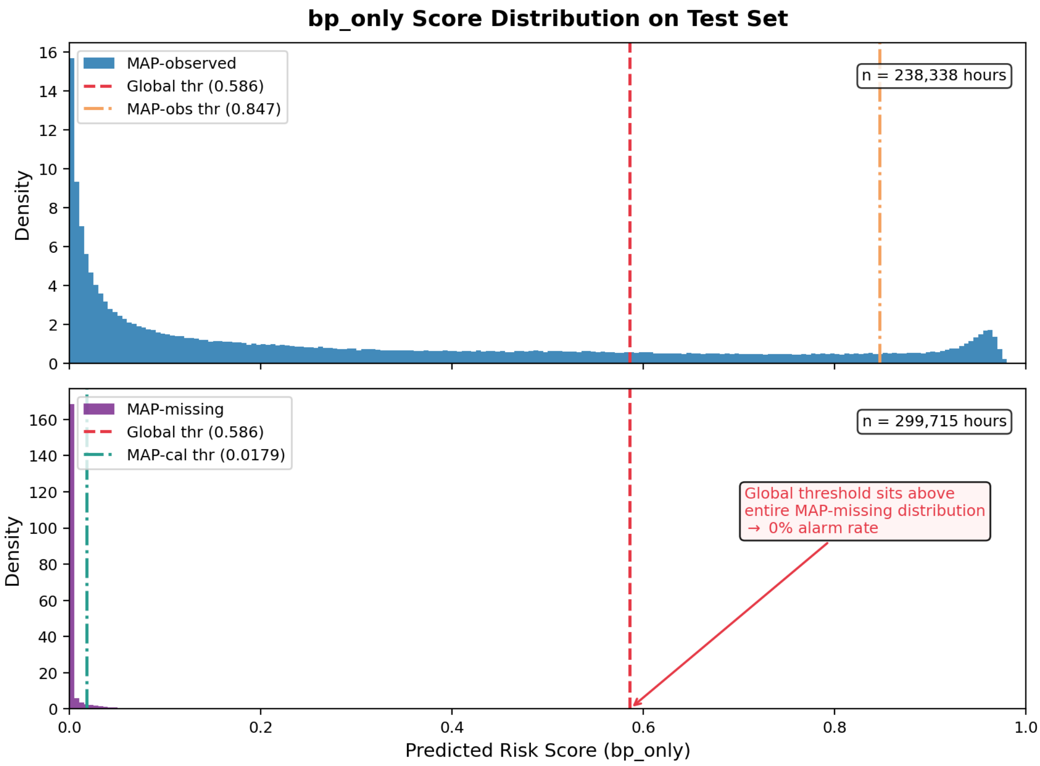


Figure 4.5: Risk score distributions for the bp_only model (compressed15, HGB) on the test set, stratified by MAP availability. **Top panel**: MAP-observed hours (238,338 hours). Scores are broadly distributed (mean 0.316, median 0.194) with a clear high-score tail extending past the global threshold at 0.586. **Bottom panel**: MAP-missing hours (299,715 hours). Scores collapse into a narrow spike near zero (mean 0.012, median 0.0001, 90th percentile 0.018). Only 0.006% of MAP-missing scores exceed the global threshold, producing an effective alarm rate of zero. The dashed red line marks the global threshold (0.586), calibrated on the full validation set to achieve a 10% alarm rate. The solid blue line marks the MAP-missing-calibrated threshold (0.018), which recovers a 10% alarm rate and 93.75% recall within the MAP-missing subset.

### 4.5.2 Subset-Calibrated Thresholding

The fix is straightforward and does not involve any changes to the model itself. Instead of applying one threshold to all hours, we calibrate a separate threshold on the MAP-missing validation subset, targeting the same 10 percent alarm rate within that subset. Because the MAP-missing scores cluster near zero, this subset-calibrated threshold is much lower: 0.018, compared to 0.586 for the global threshold.

Only 32 hypotension events in the test set fall entirely within MAP-missing hours. With so few events, all recall figures on this subset carry wide uncertainty. A 95% Clopper-Pearson confidence interval [44] for 30 out of 32 detections spans [79.2%, 99.2%], meaning the true MAP-missing event recall could plausibly be as low as 79% or as high as 99%. With that caveat, the direction of the result is clear: the adjusted threshold raises bp only's MAP-missing recall from 0.0% (0 out of 32) to 93.75% (30 out of 32). The alarm rate on the MAP-missing subset is 10.1 percent, matching the target. Table 4.8 summarises bp only performance under the global versus MAP-missing-calibrated threshold. Because only 32 MAP-missing events exist in the test set, the recall estimates in this table should be interpreted as directional rather than precise; the 95% confidence interval for the calibrated recall of 93.75% is [79.2%, 99.2%].

To assess the sensitivity of the MAP-missing recall estimate to threshold variability, a bootstrap analysis was performed. The MAP-missing validation subset (298,582 hours) was resampled with replacement 1,000 times, and for each resample a fresh 90th-percentile threshold was calibrated. Each bootstrapped threshold was then applied to the fixed MAP-missing test subset (299,715 hours, 32 events). The median bootstrapped threshold was 0.0179 (IQR 0.0178–0.0181), virtually identical to the point estimate of 0.0179. All 1,000 resamples produced the same event recall of 93.75% (30 out of 32 events). This stability arises because the validation subset contains nearly 300,000 hours, making the 90th-percentile threshold robust to resampling variation. The directional finding,that subset-calibrated thresholding recovers strong event coverage from near-zero under the global threshold, is therefore not an artefact of threshold instability. The remaining uncertainty in the recall estimate is attributable entirely to the small number of MAP-missing events in the test set ($n = 32$), not to the calibration procedure. (See Figure 4.6)

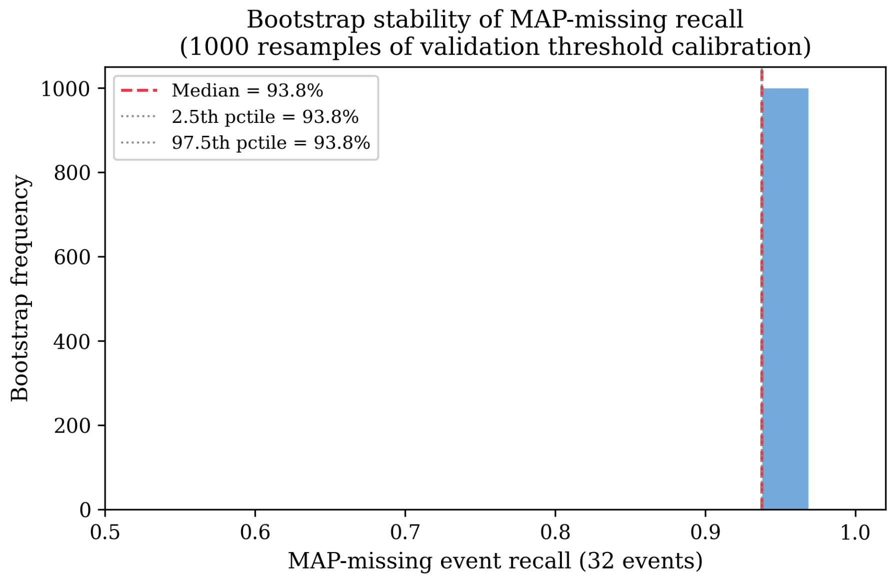


Figure 4.6: Bootstrap stability of MAP-missing threshold calibration. The MAP-missing validation subset (298,582 hours) was resampled 1,000 times with replacement; each resample produced a calibrated threshold at the 90th percentile, which was applied to the fixed MAP-missing test subset (32 events). All 1,000 resamples yielded 30 of 32 events detected (93.75% recall). The degenerate distribution reflects the large validation sample size ($n \approx 300{,}000$ hours), which makes the quantile-based threshold effectively invariant to resampling.

What makes this work? The model hasn't changed. The scores haven't changed. The

Table 4.8: bp_only model performance on the MAP-missing test subset (299,715 hours, 32 positive hours, 32 events) under the global threshold versus the MAP-missing-calibrated threshold.

| Threshold | AlarmRate | Recall | EvRecall | EvMiss |
|---|---|---|---|---|
| Global (0.586) | 0.00% | 0.0% | 0.0% | 100.0% |
| MAP-missing calibrated (0.018) | 10.1% | 93.75% | 93.75% | 6.25% |

ranking information was there all along. The measurement-timing features (hours_since_MAP, MAP_observed, MAP_count_6h) still carry valid values even when MAP itself is absent, and as Section 4.3.3 showed, hours_since_MAP is the third most important feature in the model. When MAP hasn't been measured for a long time, that feature takes a large value, which the model learned to associate with elevated risk. The problem was never the model's ranking ability on MAP-missing hours. The problem was a threshold designed for a different score distribution.

### 4.5.3 Non-BP Performance on MAP-Missing

A question is whether a model trained entirely without blood pressure features would perform better on MAP-missing hours, since it wouldn't suffer from the score distribution shift at all.

Under the global threshold, all_non_bp achieves an alarm rate of 6.1 percent on MAP-missing hours (below the 10 percent target, because its scores are also lower than the full-population average) and a recall of just 12.5 percent. With a MAP-missing-calibrated threshold, recall improves to 18.75 percent (catching 6 out of 32 MAP-missing events). The gated policy in Section 4.7 reports 25% (8 of 32) because it allocates a separate 10% alarm budget to the MAP-missing subset independently, whereas here the global threshold is applied directly, producing a lower effective alarm rate of 6.1% on that subset. That's better than nothing, but it's far below the 93.75% achieved by bp only with a calibrated threshold.

This confirms what the ablation results in Section 4.4 already suggested: non-BP features don't carry enough signal to reliably predict a MAP-defined event. The information ceiling at approximately 0.75 AUROC on the full cohort translates into even weaker performance on the MAP-missing subset, where the model must predict hypotension without ever seeing any blood pressure data for those hours. Table 4.9 shows MAP-missing performance across all auto-suite variants under both global and calibrated thresholds.

| | Global Threshold | | MAP-Missing Calibrated | | | |
|---|---|---|---|---|---|---|
| Variant | AlarmRate | Recall | AlarmRate | Recall | EvRecall | EvMiss |
| bp_only (primary) | 0.00% | 0.0% | 10.1% | 93.75% | 93.75% | 6.25% |
| bp_only (legacy) | 0.00% | 0.0% | 10.1% | 93.75% | 93.75% | 6.25% |
| all_non_bp | 6.09% | 12.5% | 9.67% | 18.75% | 18.75% | 81.25% |
| labs_only | 6.63% | 21.9% | 9.90% | 31.25% | 31.25% | 68.75% |
| pressors_dose_only | 9.32% | 25.0% | 9.89% | 25.00% | 25.00% | 75.00% |
| pressors_only | 9.43% | 18.75% | 10.06% | 21.88% | 21.88% | 78.12% |
| vitals_only | 7.29% | 6.25% | 10.11% | 12.50% | 12.50% | 87.50% |
| vitals_labs_only | 6.21% | 9.38% | 9.83% | 18.75% | 18.75% | 81.25% |
| resp_support_only | 6.56% | 6.25% | 10.08% | 12.50% | 12.50% | 87.50% |
| vent_settings_only | 8.64% | 9.38% | 10.07% | 12.50% | 12.50% | 87.50% |
| sedation_only | 9.39% | 12.5% | 10.04% | 12.50% | 12.50% | 87.50% |
| acid_base_only | 9.78% | 15.62% | 10.63% | 15.62% | 15.62% | 84.38% |
| infection_proxy_only | 9.78% | 12.5% | 10.09% | 15.62% | 15.62% | 84.38% |
| cbc_inflammation_only | 10.23% | 6.25% | 10.65% | 9.38% | 9.38% | 90.62% |

Table 4.9: MAP-missing subset performance for all auto-suite variants (299,715 hours, 32 positive hours, 32 events in the test set). The left columns show performance under the global threshold, which was calibrated on the full validation population. The right columns show performance under a threshold calibrated specifically on the MAP-missing validation subset to target a 10% alarm rate. Confidence intervals are wide due to the small number of MAP-missing events (n=32).

### 4.5.4 MAP Presence Performance Gap

Even after calibration, there's a real gap between how well the model performs on MAP-observed versus MAP-missing hours. On MAP-observed hours, bp_only achieves near-perfect event detection (over 99% event recall with the full feature set). On MAP-missing hours, the calibrated threshold recovers to 93.75% (strong, but not equivalent). Table 4.9 show both subsets side by side. And the MAP-missing subset contains only 32 events in the test set, so the confidence interval around that 93.75% estimate is wide. With so few events, the difference between catching 30 and catching 28 would move the number by 6 percentage points.

This gap is partly calibration (solved by the subset threshold) and partly structural. When MAP is absent, the model relies on indirect signals: measurement timing, heart rate patterns, lab recency, ICU unit type. These carry real information (the permutation importance analysis confirmed that) but they are noisier and less specific than direct blood pressure readings. No amount of threshold adjustment can fully compensate for having less informative inputs. The small event count (n = 32) also means the gap itself is imprecisely estimated, therefore a multi-site study with a larger MAP-missing event population would be needed to determine the stable performance difference between the two regimes.

### 4.5.5 Deployment Interpretation

The practical implication is clear: any alarm system built primarily on blood pressure features must explicitly handle the regime where blood pressure isn't being measured. A globally calibrated threshold will silently fail for more than half of all patient-hours in a typical ICU EHR dataset, and this failure is invisible in standard aggregate evaluation. It only shows up when you stratify.

The approach evaluated in this thesis, applying a lower, separately calibrated threshold to MAP-missing hours (what we call Config G, formally introduced in Section 4.7), recovers strong event coverage at the cost of a modestly higher overall alarm rate. Config G produces an overall event recall of 97.6% with approximately 0.89 alerts per 24 patient-hours under a six-hour cooldown. The MAP-missing hours receive alarms at a 10% rate matched to MAP-observed hours, rather than the 0% they would receive under a global threshold.

Alternative approaches exist. A gated policy could switch between the bp_only model for MAP-observed hours and a non-BP fallback model for MAP-missing hours. But as Section 4.5.3 showed, the non-BP model only catches 18.75% of MAP-missing events even with a calibrated threshold, compared to 93.75% for BP only with its timing features. So the single-model dual-threshold approach outperforms the two-model gated approach by a wide margin on this dataset. The measurement-timing features in BP only are simply more informative than the full set of non-BP physiological signals.

## 4.6 Ablation Suite Comparisons

### 4.6.1 Full Cohort and MAP-Missing Rankings

Table 4.5 ranks all 15 auto-suite variants by test AUROC. bp_only sits at the top with an AUROC of 0.945, and there is a gap of 0.165 before the next variant (all_non_bp at 0.780). Below that, the non-BP variants form a cluster between 0.617 and 0.780, with the ordering roughly following clinical intuition, composites that combine multiple signal families (all_non_bp, vitals_labs_only) outperform the single-block isolates, and blocks with more direct physiological relevance to hemodynamics (labs, respiratory support, vitals) outperform those with weaker or more indirect connections (CBC inflammation, acid-base, infection proxy).

However the full-cohort ranking does not tell the whole story, when looking at the MAP-missing performance (Table 4.9), the picture changes. bp_only model goes from first place to a special case: it scores zero under the global threshold but recovers to 93.75% recall after calibration. Among the non-BP variants, labs_only becomes the strongest standalone block for MAP-missing recall, with it being at 31.25%, much ahead of all_non_bp model at 18.75% and pressors_dose_only at 25%. This reversal, with the labs_only model outperforming the full non-BP composite model on MAP-missing hours is likely a small sample artifact (n = 32 events), but it shows that adding more features does not automatically help in a low signal feature regime.

Single-block variants with the weakest full-cohort performance (cbc_inflammation_only at 0.617, acid_base_only at 0.637) also have the weakest MAP-missing recall (9.4% and 15.6% respectively). These blocks capture metabolic signals that are too sparse to provide an useful signal on their own, regardless of MAP availability.

The comorbidity_static_only variant was skipped as it has constant features across all hours within a stay, therefore no pattern to learn from.

### 4.6.2 Cross Comparison

In Table 4.6, two findings stand out. First, among BP-containing variants, blood pressure features account for nearly all of the predictive signal. The bp_only model (37 features) achieved an AUROC of 0.955 and an event miss rate of 0.66%. The bp_plus_gcs (46 features) and bp_plus_resp_support (71 features) variants were essentially tied on AUROC and showed very similar event-level performance, indicating that small additions to blood pressure features do not yield a clear practical gain. In contrast, larger expansions such as bp_plus_labs (154 features) and bp_plus_all_new (308 features) reduced performance, suggesting that adding broader feature blocks can introduce noise without improving discrimination.

Second, that among non-BP variants, the information ceiling is quite flat. vitals_labs_only scores 0.746, vitals_labs_pressor_fluids scores 0.746, vitals_labs_gcs scores 0.746, vitals_labs_resp_support scores 0.746, the full five-block composite scores 0.748, and all_non_bp scores 0.747. Adding respiratory support, GCS, sedation, vasopressor dose, infection proxies, and comorbidity indices on top of vitals and labs buys less than 0.002 AUROC. The feature set went from 161 features to 319 features and gained nothing. This isn't a model capacity problem as HGB with 12,000 trees has plenty of capacity. It's a signal problem. Without blood pressure, the available clinical signals simply plateau at around 0.75 AUROC for predicting a MAP-defined event.

The cross-suite comparison (Table 4.7) reveals the cost of feature simplification but should be interpreted with caution because two things change between suites at once: the backend (auto-selected XGBoost or HGB in the auto-suite versus HGB only in compressed15) and the feature set (simplified versus full). This means performance differences between suites cannot be cleanly attributed to either factor alone.

With that caveat, the bp_only comparison is informative because the backend happens to differ while the signal block is the same. The full feature set (compressed15) outperforms the simplified set (auto-suite) by 1 percentage point in AUROC (0.955 vs 0.945) and over 5 percentage points in recall (87.8 vs 82.2 percent). The dropped features (baseline medians, deltas, rolling min/max/std) do carry useful information for the blood pressure block. For non-BP variants, the auto-suite XGBoost slightly outperforms compressed15 HGB (0.780 vs 0.747 for all_non_bp), but we cannot determine whether this reflects the backend, the feature set, or both. What we can say is that the non-BP ceiling sits somewhere between 0.747 and 0.780 depending on configuration, and neither number approaches the 0.955 achieved by bp_only model. One possible explanation for the auto-suite's slightly higher non-BP AUROC is that feature simplification reduces overfitting on sparse signals. Non-BP features are noisier and more intermittent than blood pressure, so a smaller feature set may allow the model to focus on the most reliable predictors rather than fitting noise in rolling statistics and baseline deviations.

## 4.7 Gated Policy Evaluation

The results in Section 4.5 showed that the global threshold produces zero alarms on MAP-missing hours and that a calibrated threshold recovers a 93.75% recall on those hours. However, that analysis took each subset in isolation. In a real deployment setting, the system would need to handle both regimes simultaneously, using one policy when MAP is observed and another when it is not. The following analysis uses the compressed15 bp_only model (HGB, 37 features), whose MAP-observed calibrated threshold is 0.847 and MAP-missing-calibrated threshold is 0.018. These differ from the auto-suite thresholds reported in Section 4.5.2 (global 0.586) because the two suites use different backends and feature sets, which produce different score distributions.

We tested six gated switching configurations (Configs A through F) and compared them against a single model dual threshold approach (Config G), hence the name.

Configs B through F use bp_only model at a 10 percent alarm rate on MAP-observed hours and all_non_bp on MAP-missing hours at progressively relaxed budgets (10, 15, 20, 25, and 30 percent). Config A is the baseline, which is bp_only with its global threshold only, which gives zero coverage on MAP-missing hours.

The outcome was counterintuitive. Event recall was 97.5% across every Configs A through F. Adding the all_non_bp fallback at any alarm budget did not improve event detection by a single event. Only 32 out of 1,053 test events (3.0 percent) fall exclusively in MAP-missing hours. Most events have at least some MAP-observed hours in the three hour pre-onset window, so bp_only can fire through those hours regardless of what happens during MAP-missing hours.

Meanwhile, the fallback model's recall on those 32 MAP-missing events was poor. At a 10% alarm rate, all_non_bp caught 8 of 32 (25%). At 20 percent, 14 (43.8%). At 30 percent, 20 (62.5%). The relationship is roughly linear with no efficient operating point. To match 93.75 percent recall, the fallback would need to alarm on 50 percent of MAP-missing hours which is approximately 150,000 alarms for 30 extra event detections.

Config G takes a different approach. It uses bp_only for both subsets but applies a different threshold depending on MAP availability: 0.847 for MAP-observed hours and 0.018 for MAP-missing hours. No second model is involved. Config G achieves 97.6 percent overall event recall, 93.8 percent MAP-missing event recall, and 0.89 alerts per 24 hours under a six hour cooldown. It outperforms every gated configuration on MAP-missing detection (93.8 versus 25.0 percent at matched alarm rate) because bp_only's measurement timing features (hours_since_MAP, MAP_observed) carry signal even when the values themselves are absent. Table 4.10 summarises all configurations.

| Config | Strategy | Overall AR | EvRecall | Miss AR | Miss EvRecall | Alerts/24h |
|---|---|---|---|---|---|---|
| A | bp_only MAP-obs thr only | 4.7% | 97.5% | 0.0% | 0.0% | 0.41 |
| B | gated 10/10 | 10.3% | 97.5% | 10.0% | 25.0% | 0.74 |
| C | gated 10/15 | 13.1% | 97.5% | 15.1% | 37.5% | 0.88 |
| D | gated 10/20 | 15.8% | 97.5% | 20.0% | 43.8% | 1.01 |
| E | gated 10/25 | 18.5% | 97.5% | 24.9% | 53.1% | 1.13 |
| F | gated 10/30 | 21.2% | 97.5% | 29.7% | 62.5% | 1.27 |
| **G** | **bp_only dual threshold** | **10.3%** | **97.6%** | **10.1%** | **93.8%** | **0.89** |

Table 4.10: Deployment configuration comparison on the test set (538,053 hours, 1,053 events, 32 MAP-missing events).

Config G is the best-performing configuration in this retrospective evaluation: one model, two thresholds, no switching logic. It outperforms the two-model gated approach on every metric tested, making it the recommended candidate for prospective validation.

## 4.8 Error Analysis and Case Studies

### 4.8.1 Example Stay Timelines

Three annotated patient trajectories are provided (Figures 4.8 to 4.10) to illustrate the alarm system's behaviour across different clinical scenarios. Each figure shows the risk score timeline (top panel), MAP and heart rate values (middle panel), and MAP observation status (bottom panel). Stay A (Figure 4.8) represents the intended use case. MAP is observed throughout the stay and declines gradually from the high 80s into the low 60s. At hour 22, three hours before onset, the score crosses the threshold and the first alarm fires.

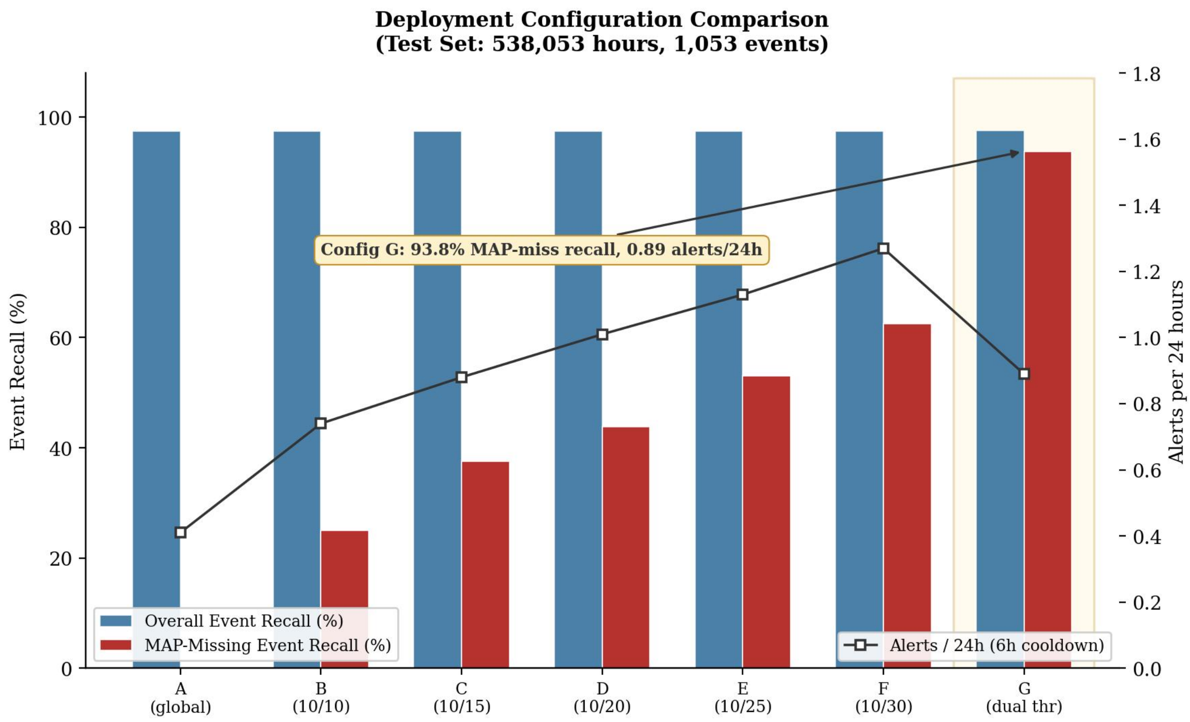


Figure 4.7: Deployment configuration comparison. Config G (one model, dual thresholds) achieves 93.8% MAP-missing event recall at 0.89 alerts per 24 hours, outperforming all gated two-model alternatives.

Alarms continue through hours 23, 24, and 25, giving clinicians the full three hour warning window. Both the global and Config G thresholds produce identical behaviour because MAP is always observed.

Stay B (Figure 4.9) is the missed event described above. It demonstrates an honest limitation of the system: patients with volatile, recovering MAP trajectories that collapse suddenly can evade detection at the 10 percent alarm budget. Lowering the threshold would catch this event but at the cost of more false alarms across the cohort.

Stay C (Figure 4.10) illustrates the core contribution most directly. MAP is observed through hour 14 (stable, in the 79 to 92 range), then disappears at hour 15. Under the global threshold (0.847), the model falls silent for the entire MAP gap and remains silent even when MAP returns at hour 20 with a value of 71, well below the pre-gap level. The global threshold never fires before onset at hour 21. It fires at hour 22, one hour after onset, which is too late to count as early warning.

Under Config G, the threshold drops to 0.018 when MAP becomes absent at hour 15. The score at that hour is 0.085, which is well above 0.018, and an alarm fires immediately. A second alarm follows at hour 16. Config G fires at hour 15, six hours before onset in wall-clock terms. Formally, this alarm falls outside the three hour prediction horizon defined in Section 3.4.3, the label window for onset at hour 21 covers only hours 18 through 21.

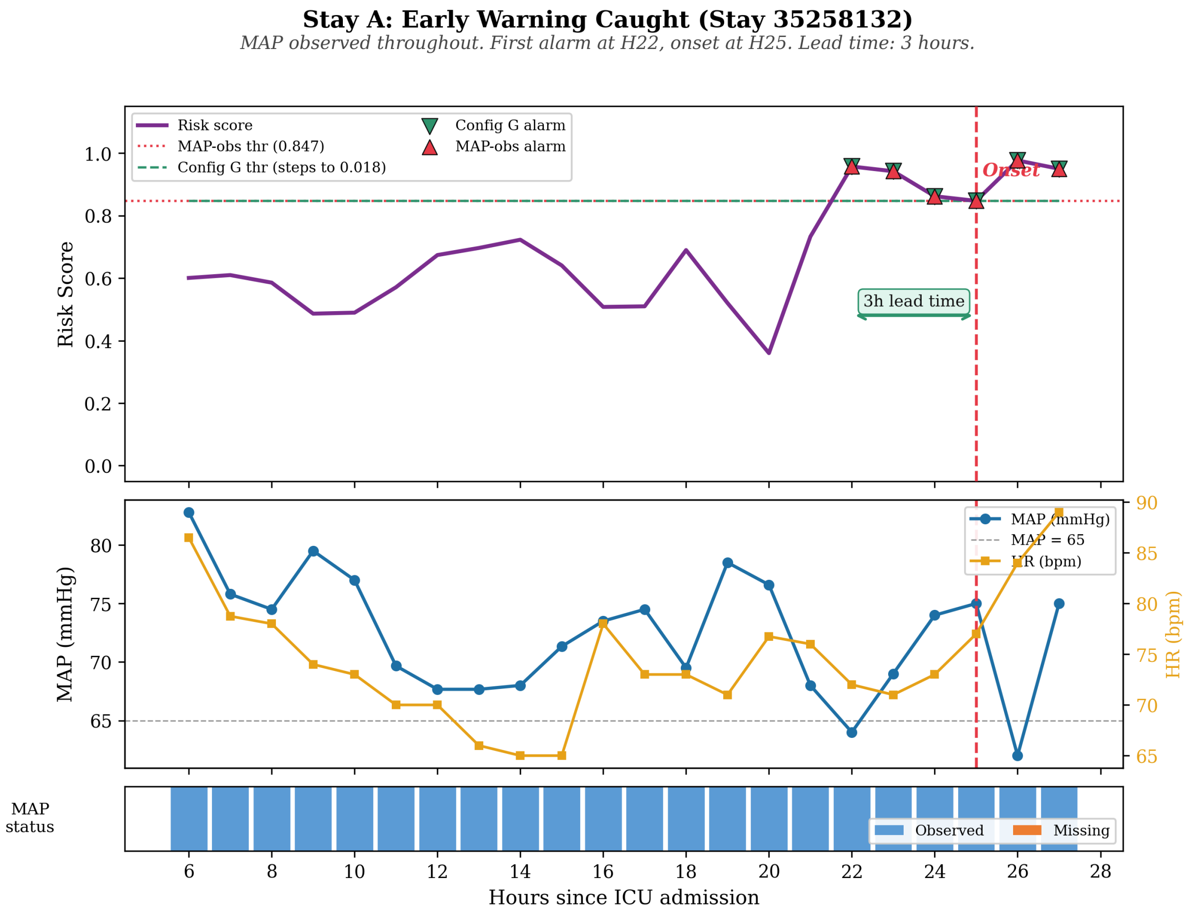


Figure 4.8: Stay A (stay 35258132): early warning caught. MAP is observed throughout. The risk score rises as MAP declines into the low 60s, crossing the 0.847 threshold at hour 22. The first alarm fires 3 hours before onset at hour 25.

The alarm at hour 15 is therefore not counted as a true positive at the hourly level, and the six hour gap is not comparable to the lead times reported in Section 4.3.3. At the event level, however, a clinician receiving this alert would have six hours of advance notice in wall-clock terms. This observation falls outside the formal three-hour evaluation window and should be interpreted as anecdotal illustration rather than quantitative evidence, but it highlights the practical value that Config G can provide during MAP-missing hours. The global threshold misses this event entirely. Same model, same scores; the only difference is which threshold is applied during the MAP-missing window. This example encapsulates the core argument, a globally calibrated threshold silently overlooks patients whose monitoring pattern produces lower scores, and a threshold adjustment recovers early warning coverage without requiring a different model.

### 4.8.2 Analysis of Missed Events

Under the compressed15 configuration, bp_only model missed 7 out of 1,053 test events (0.66 percent event miss rate). Under Config G's dual threshold policy, 2 out of 32

MAP-missing events were also missed, giving 93.75 percent event recall on that subset. What characterises the events that the model fails to catch?

Missed events were examined by extracting their full hourly trajectories from the test set. A consistent pattern emerged. The missed events usually involved patients whose MAP fluctuated in the borderline range (65 to 75 mmHg) for extended periods, with repeated partial recoveries that pulled the risk score back down before onset. In these cases, the model correctly reduced its score when MAP bounced back, which is the right behaviour for avoiding false alarms during transient dips. But when the final deterioration came, it arrived abruptly from a seemingly recovering trajectory, and the model's score did not cross the threshold before the onset hour.

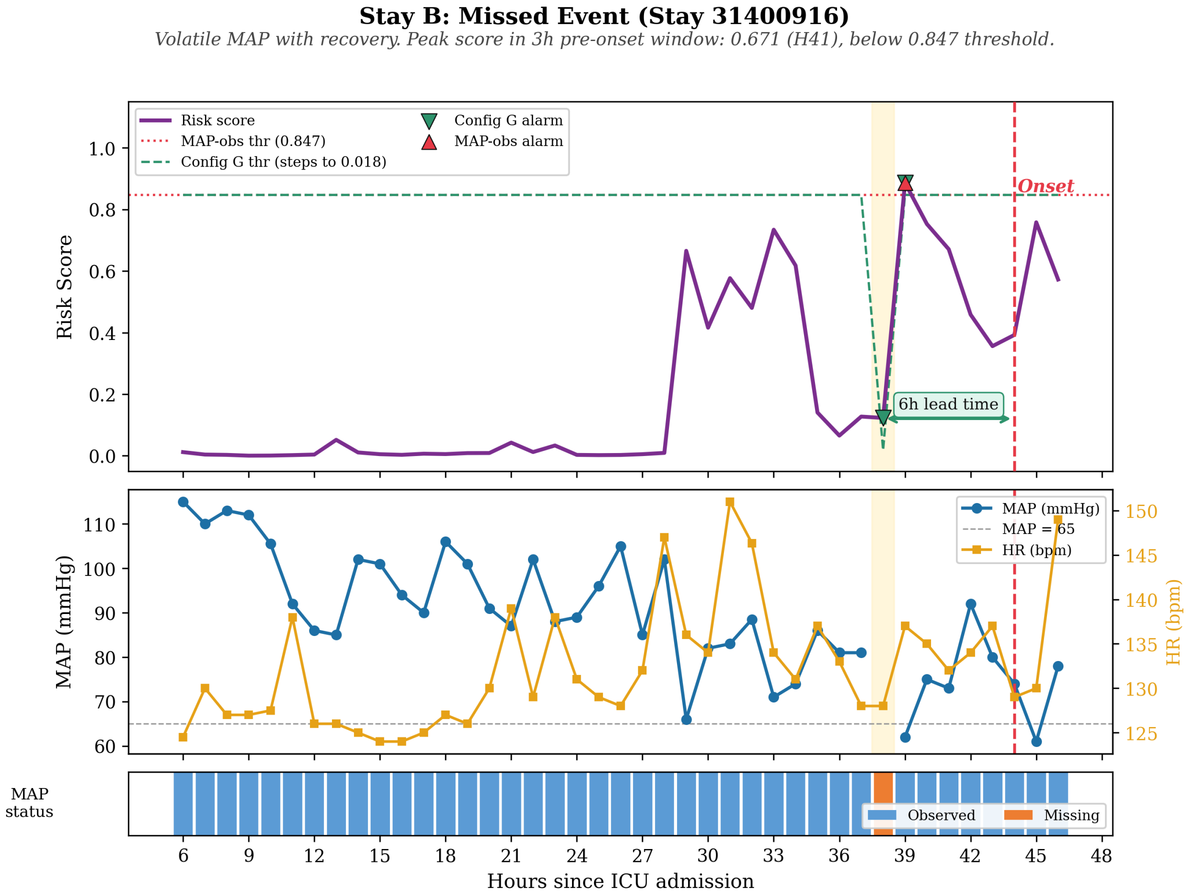


Figure 4.9: Stay B (stay 31400916): missed event. MAP fluctuates between 62 and 115 with repeated recoveries. The model fires at hour 39 (MAP = 62, score = 0.887) but this falls outside the 3-hour pre-onset window

Stay B illustrates this pattern. The patient's MAP oscillated between 62 and 115 with repeated recoveries and no sustained downward trend. The highest risk score within the three hour pre-onset window was 0.671 at hour 41, below the 0.847 threshold. The model did fire at hour 39 (MAP of 62, score of 0.887) but this fell outside the three hour window before onset. The model's peak score within the relevant window fell roughly 21% below the

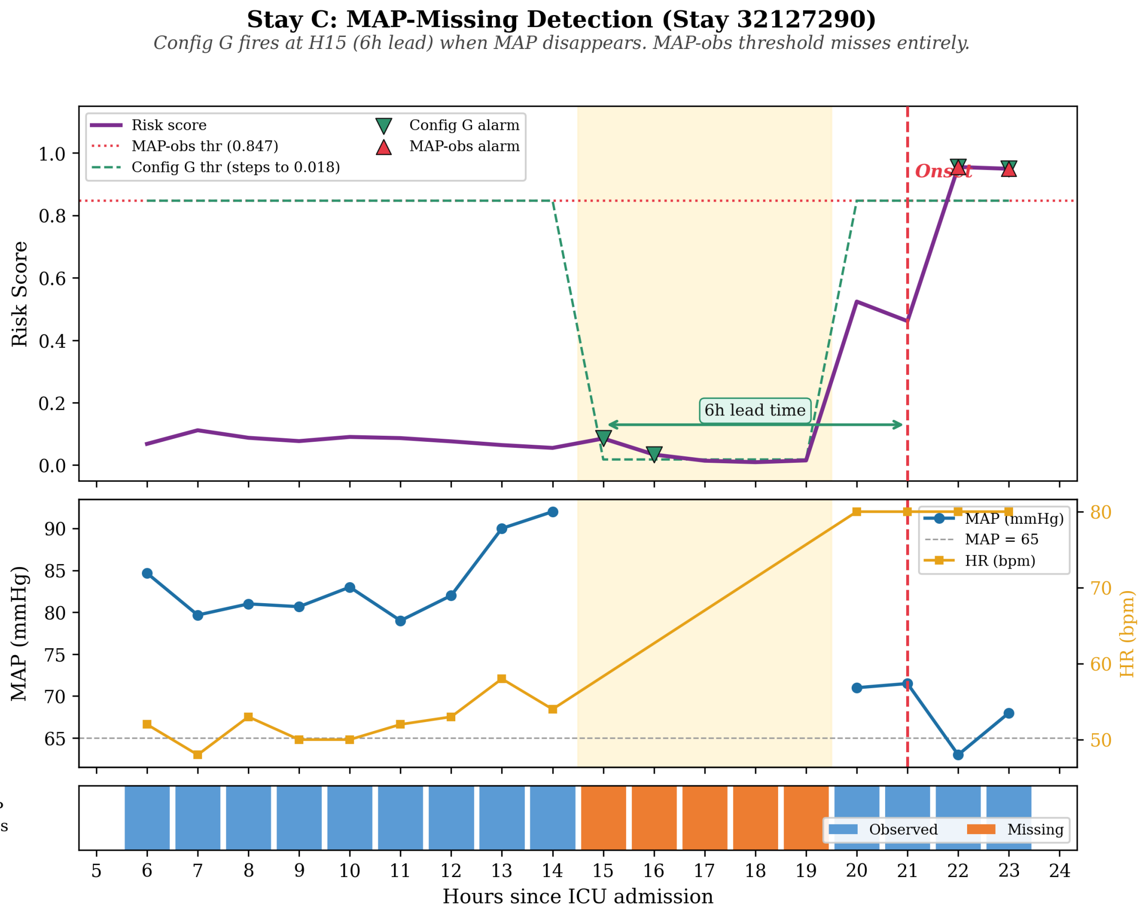


Figure 4.10: Stay C (stay 32127290): MAP-missing detection with Config G. MAP is stable (79 to 92) through hour 14, then disappears at hour 15 (amber shading). Under the MAP-observed threshold (0.847), no alarm fires before or after the gap. Under Config G, the threshold drops to 0.018 during MAP-missing hours, and alarms fire at hours 15 and 16 (green markers), six hours before onset at hour 21 in wall-clock time. This falls outside the formal three hour label window but provides practical advance warning that the global threshold misses entirely

threshold.

This type of miss reflects a tradeoff. A lower threshold would have caught Stay B but would also have generated more false alarms on the many patients who show similar borderline oscillations and never develop sustained hypotension. At the population level, the model's calibration is correct: 99.3 percent of events are caught. The 0.7 percent that slip through tend to be sudden collapses from volatile but non-trending trajectories.

## 4.9 Summary of Results

Five findings define the experimental contribution of this thesis.

First, blood pressure features dominate hypotension prediction at the hourly level. The bp_only model achieved an AUROC of 0.955 and caught 99.3 percent of events in the compressed15 suite using just 37 features. Small additions such as GCS or respiratory support produced broadly comparable performance, but larger additions did not improve results and in several cases slightly degraded them. bp_plus_all_new (308 features) scores lower than bp_only on every metric.

Second, non-BP features plateau at an information ceiling around 0.75 AUROC regardless of how many signal families are included. Six progressively richer non-BP composites (from 161 to 319 features) all scored between 0.746 and 0.748 in the compressed15 suite. This is not a model capacity limitation or a feature engineering gap but it reflects a genuine signal boundary. Without direct blood pressure measurements, the available clinical signals cannot predict a MAP-defined event with the same accuracy.

Third, the global alarm threshold generates zero alarms for 55.7% of test hours. When MAP is absent, the bp_only model's scores fall into a narrow band near zero (mean is 0.012, 90th percentile 0.018), and the global threshold (0.586) sits above 99.994% of those scores. All 32 MAP-missing events in the test set receive no warning under this threshold. Standard aggregate metrics (AUROC, event recall on the full cohort) hide this failure entirely.

Fourth, a subset-calibrated threshold on the same model recovers 93.75% recall on MAP-missing hours ($n = 32$ events; 95% CI: 79.2%–99.2%). Bootstrap analysis confirmed the threshold calibration is stable (Section 4.5.2), the width of the confidence interval reflects the small event count, not calibration noise. The model's ranking ability was intact all along as measurement timing features (hours_since_MAP ranked third in permutation importance) carry predictive signal even when MAP values are absent.

Fifth, Config G (one model, two thresholds) is the strongest configuration tested in this retrospective evaluation. Its primary advantage is MAP-missing event recall (93.8% versus 25.0% for the best gated alternative at matched alarm rate), while overall event recall is comparable across configurations (97.5–97.6%). It achieves this at 0.89 alerts per 24 hours under a six-hour cooldown and outperforms all gated two-model policies, which max out at 25 percent MAP-missing event recall at matched alarm rates because the all_non_bp fallback lacks the timing features that BP only learned during training.

These results raise several questions.

- Why measurement timing carries such strong predictive signal,
- how the 32 MAP-missing events should be interpreted given the wide confidence intervals at that sample size,
- what the non-BP ceiling means for the limits of EHR-based prediction, and
- how the findings relate to prior work on hypotension forecasting.

These are addressed in the next chapter.

# 5 Discussion

## 5.1 Summary of Main Findings

This study aimed to build an hourly early warning system for sustained hypotension in ICU patients and to test whether that system could maintain useful performance when its primary input signal, Mean Arterial Pressure, was unavailable. Five findings emerged: blood pressure features dominate at the three hour horizon (AUROC 0.955, 99.3% event recall with 37 features); non-BP features plateau at an information ceiling around 0.75 AUROC in which a globally calibrated threshold produces zero alarms for 55.7% of test hours where MAP is absent; subset-calibrated thresholds on the same model recover 93.75% recall without retraining, and Config G (one model, two thresholds) achieves 97.6% event recall at 0.89 alerts per 24 hours in this retrospective evaluation. The following sections examine why these patterns arise, what they imply for deployment, and where the analysis falls short.

## 5.2 BP Dominates at a Three Hour Horizon

The dominance of blood pressure features is not surprising once you consider the structure of the prediction task. Sustained hypotension is defined by MAP falling below 65 mmHg for at least two out of three consecutive hours. The label is literally a function of the feature, so it is not surprising that the model excels here. So a model that receives current MAP, rolling MAP statistics over the preceding six hours, and the rate of change from baseline has direct access to the signal that determines the outcome. The model is being asked to predict whether a downward MAP trajectory will continue and has the trajectory itself as input.

At a three hour horizon, this direct path is especially strong. MAP dynamics over the preceding hours are temporally closer to the prediction target, so recent MAP values are highly correlated with near future MAP behaviour. Longer horizons would weaken this correlation because more can change between the prediction point and the event. Shorter horizons would strengthen it further but leave less time for a clinical response. Three hours appears to be a horizon where MAP dynamics are still informative enough to achieve strong discrimination while providing a clinically useful warning window.

Beyond just the values, the bp_only model also benefits from having measurement pattern features, hours_since_MAP ranked third in permutation importance, ahead of most direct value features. This is consistent with the informative observation argument made in Section 2.3.3 [29], when a patient's MAP is being measured frequently, clinicians likely suspect instability, , and that clinical concern correlates with higher risk. When MAP has not been measured for several hours, the model still receives a meaningful input (a large hours_since_MAP value) rather than just absence. This is also why Config G works. The measurement timing features remain populated even when MAP values are missing, giving the model something to rank on. The subset-calibrated threshold then translates that ranking into alarms at the correct operating point for the MAP-missing score distribution. Without these timing features, the bp-only model's scores during MAP-missing hours would carry no useful ordering, and subset calibration alone would not recover meaningful recall.

Nakanishi et al. reached a similar conclusion using a different architecture and data resolution [20]. In their study, an LSTM trained on BP-only features at minute-level resolution outperformed the same architecture trained on all available features. This suggests that BP dominance for hypotension prediction is not an artefact of our particular feature engineering or model choice but it reflects the structure of the task itself.

## 5.3 Why Non-BP Signals Plateau

Given that non-BP signals achieved only 0.747 AUROC on their own, the natural question is why adding them on top of blood pressure did not help either. After all, heart rate, respiratory rate, lactate, and other clinical signals do carry real information in ICU populations. Several factors explain the limited incremental value.

Non-BP signals are physiologically related to hemodynamic instability but, as discussed in Section 2.3.1, are less specific at an hourly resolution window. Heart rate rises in response to fall in blood pressure via the baroreceptor reflex, but it also rises in response to fever, pain, anxiety and many other stimuli [3]. Elevated lactate can reflect tissue hypoperfusion, but results arrive intermittently and often lag behind the event by hours. So by the time a lab result confirms what MAP already showed, the model has already since made its prediction [25].

The non-BP signals are also noisier. SpO2 and respiratory rate fluctuate with patient movement, suctioning and ventilator changes. Vasopressor and sedation features are mostly reactive than predictive (meaning vasopressors are given in response to hypotension, so the signal often happens post-event), including these features can shift the model's score distribution without adding information, which explains why models like bp_plus_labs and bp_plus_all_new both slightly degraded performance relative to the bp_only model.

The information these signals contain may already be captured indirectly by the blood pressure trajectory. A patient with declining MAP over six hours and increasing measurement frequency likely has elevated heart rate, and the model can infer this from the MAP dynamics and timing features itself. Adding heart rate as a separate feature is not adding much new information.

However, this does not mean non-BP features are useless as their value appears in the MAP-Missing regime, where they provide the only available signal. all_non_bp achieved 0.780 AUROC on the full cohort (auto-suite) , considerably better than chance. Cherifa et al. similarly reported an AUROC of approximately 0.83 using non-BP features at five minute resolution [21], and Moghadam et al. found that removing blood pressure features from their ICU hypotension model substantially reduced performance [19], consistent as with our finding. However these signals are not additive to blood pressure in the way one may hope. For a MAP-defined outcome at this horizon and within this dataset, MAP appears both necessary and nearly sufficient.

It is worth noting that this ceiling may be specific to the prediction target. Sustained hypotension is defined directly in terms of MAP, so predicting it without MAP is analogous to predicting a lab result without access to that lab value. For deterioration targets defined by broader physiological criteria, such as composite organ failure scores or clinical deterioration requiring escalation of care, non-BP signals might contribute more because the outcome would no longer be a function of a single variable. Whether the 0.75 AUROC ceiling persists for other ICU prediction tasks is an open question that this study cannot answer but that would be valuable to test.

## 5.4 MAP-Missing Coverage Disparity

The most consequential finding in this thesis is not just the model result but also the evaluation result, which is that a globally calibrated alarm threshold renders the alarm system completely silent for approximately 56% of patient-hours. This failure is not caught under standard reporting practices because, as Section 2.3.3 established, aggregate metrics average MAP-observed and MAP-missing performance into a single number. An AUROC of 0.955 and an event recall of 99.3% look good on paper but conceal the fact that 299,715 hours receive a risk score but no alarm, as Figure 4.5 makes visually clear, and 32 hypotension events pass without any warning.

Two distinct problems are at work. First is a calibration problem as the global threshold is

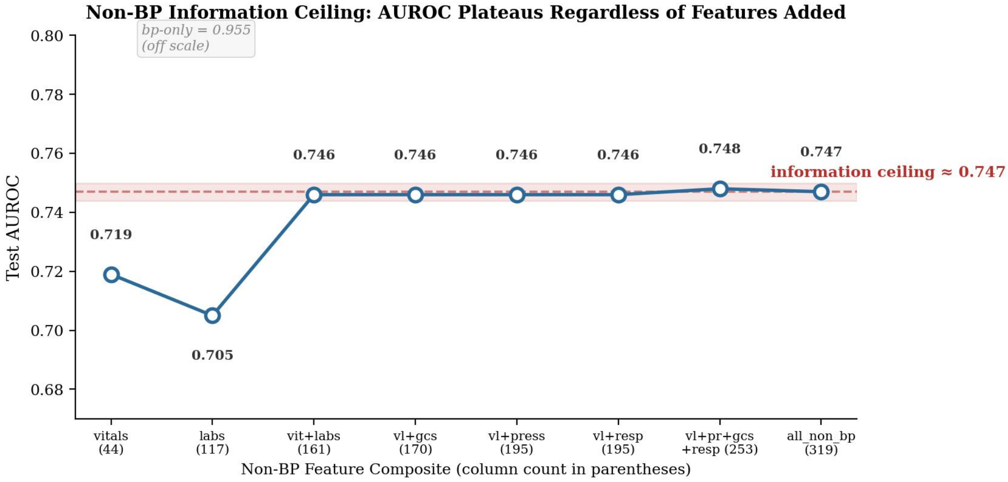


Figure 5.1: Non-BP information ceiling. Adding feature blocks beyond vitals and labs produces no meaningful AUROC improvement. Six composites from 161 to 319 columns all score between 0.746 and 0.748.

set at the 90th percentile of the full validation score distribution, which mixes different populations. MAP-observed hours produce scores spread broadly (mean 0.316). A single threshold cannot serve both regimes at a 10% alarm rate. This problem has a clean fix though, which is to calibrate separate thresholds for each subset, as the proposed model does. After recalibration, the bp_only model recovers 93.75% recall on MAP-missing hours using the same model and the same scores.

The second problem is structural. Even after recalibration, predicting hypotension without blood pressure data is harder as non-blood-pressure-related signs carry weaker and noisier evidence. all_non_bp model reaches only 18.75% recall on MAP-missing hours at 10% alarm rate. Subset calibration helps the blood pressure only (bp_only) model because its timing features (hours_since_MAP, MAP_observed) retain predictive signal during MAP gaps, but the prediction still is not as precise as when MAP values are available. The 93.75% figure comes from only 32 events, and bootstrap resampling of the threshold calibration confirmed that this recall estimate is stable across 1,000 resamples (Section 4.5.2). The remaining imprecision is therefore attributable to test-side discreteness rather than calibration noise.A 95% Clopper–Pearson confidence interval for 30 out of 32 is [79.2%, 99.2%], so the true recall could plausibly be as low as 79%. Catching 28 out of 32 would drop the point estimate to 87.5%.

For deployment, this implies that any alarm system built on blood pressure features must explicitly account for hours when blood pressure is not measured as a single threshold will not work. This pattern is likely not specific to our model or dataset. Any setting where a BP-based early warning system operates on intermittently charted EHR data would be expected to face a similar distributional shift between monitored and unmonitored hours, though the severity would depend on local MAP missingness rates and charting practices. The fix is model-agnostic and requires no additional infrastructure, just a threshold lookup conditioned on whether MAP was observed in the current hour. Figure 4.7 shows that Config G outperforms every gated alternative on MAP-missing event recall while maintaining a lower alert burden.

No prior hypotension study we identified, including the directly comparable OneICU baseline on MIMIC-IV hourly data, stratifies evaluation by MAP availability [7]. OneICU explicitly excluded windows with missing MAP from their evaluation. Nakanishi et al. and Cherifa et al. similarly report aggregate metrics without this subset stratification [20, 21]. The absence of this analysis in literature is due to standard evaluation frameworks overlooking it.

## 5.5 Event-Level vs Hour-Level Evaluation

One pattern that recurs throughout the results is the gap between the hour-level and the event-level performance. The bp_only model catches 82.2% of positive hours but 98.3% of events. So how can a model miss 18% of positive hours yet catch 98% of events?

Since hypotension events span multiple hours, a sustained event with onset at hour 25 generates positive labels at hours 22, 23, 24 and 25 (within the 3 hour horizon). If the model fires an alarm at hour 22 and hour 23 but does not fire at hour 24, it missed one positive hour but still catches the event. Clinically, this is the right way to evaluate (a clinician needs at least one warning before deterioration, not a continuous alarm).

This distinction also explains why the coverage disparity has a smaller event-level impact than the hour-level numbers suggest. MAP is missing for 55.7% of test hours, but only 32 out of 1,053 events (3 percent) fall entirely within MAP-missing windows. Most events have at least some MAP-observed hours in the pre-onset period, allowing bp_only to fire through those hours even if other hours in the window lack MAP. Config A (bp_only with global threshold, zero coverage on MAP-missing hours) still achieves 97.5% event recall because the model catches events through their MAP-observed portions.

It is recommended that future ICU early warning studies report event-level metrics alongside hourly metrics as standard practice, since event recall and event miss rate more directly reflect clinical utility than hour-level recall alone [13].

## 5.6 Methodological Implications

Three methodological choices made in this study have a broader implication for ICU prediction research.

First, the event definition had to be adapted to the data. Strict consecutiveness (requiring low MAP in consecutive hours) produced zero events in MIMIC-IV because MAP is charted too sparsely for consecutive-hour evidence to accumulate reliably. The window-based persistence proxy (two out of three hours) preserves the clinical intent of sustained hypotension events while being tolerant to documentation gaps that are inherent in non-waveform EHR data. This adaptation is a necessary response to the data reality described in Section 2.3.2 . Other non-waveform ICU prediction studies faces the same challenge and resolve it with similar compromises [20, 21], though the specific rules differ across the studies, which limits cross-study comparability [7].

Second, the fixed alarm budget framework proved essential for getting performance differences that ranking metrics alone would hide. bp_only and all_non_bp both produce well-separated score distributions (AUROCs of 0.955 and 0.780 respectively), but the operational gap is much larger than the AUROC difference suggests. At a 10% alarm rate, bp_only catches 99.3% of events while all_non_bp model catches 39.2% . AUROC integrates over all thresholds, the alarm budget forces a comparison at the threshold that matters for deployment [12, 13].

Third, observation-pattern features validated the informative missingness hypothesis from Section 2.3.3. hours_since_MAP ranked third in permutation importance for bp_only, and timing features accounted for five of the top 10 features in the all_non_bp model. Measurement timing is not noise that is to be imputed but a signal that the model exploits [22, 29]. For ICU prediction studies that currently discard or impute missingness patterns, these results suggest that encoding when and how often variables are measured can meaningfully improve both discrimination and interpretability.

## 5.7 Limitations

All experiments use a single dataset from a single institution. MIMIC-IV reflects charting practices, patient populations, and treatment protocols at Beth Israel Deaconess Medical Center between 2008 and 2022 [6]. Hospitals with different documentation workflows, monitoring equipment, or patient demographics may produce different MAP missingness rates, different feature distributions, and different model performance. The core finding (that global thresholds fail when MAP is absent) should generalise to any setting where MAP is charted intermittently, but the specific thresholds, recall numbers, and feature importance rankings are tied to this dataset.

This study uses only non-waveform charted data. Hospitals with continuous arterial line monitoring would have far lower MAP missingness rates, and the threshold collapse problem would affect a smaller fraction of patient-hours. Waveform-based systems operating at second or minute-level resolution would also likely outperform hourly models on short horizon prediction, because they observe MAP dynamics at finer temporal granularity [16]. The results are for a setting where continuous waveform data is unavailable, which is common but not universal.

The hypotension label is an operational proxy, not a clinician-adjudicated outcome. The persistence rule (two out of three hours with MAP below 65) was designed to capture sustained hemodynamic instability under the sparse charting conditions described in Section 2.3.2, but it has not been validated against clinical chart review. Some labelled events may represent transient measurement artefacts, and some genuine hypotension episodes may be missed because MAP was not charted during the relevant hours. The label reflects what the data can support, not necessarily what a clinician would identify.

Finally, the observation window is restricted to the first 48 hours of each ICU stay. Patients who develop hypotension later in their admission are not captured, and the feature baselines computed over hours 0 through 5 become less meaningful for predictions at later time points. Extending the window would increase cohort size and capture additional events, but would also introduce confounding from longer-stay care patterns that differ from the acute admission phase.

The MAP-missing subset of the test set contains only 32 hypotension events. All recall figures reported for this subset carry wide confidence intervals; the difference between catching 30 and catching 28 events would shift recall from 93.75% to 87.5%. While the direction of the finding (that subset calibration recovers strong recall from zero) is robust, the precise recall estimate should not be treated as stable. A larger dataset or a multi-site study with higher MAP-missing event counts would be needed to narrow this uncertainty.

There is also a potential selection bias in the event definition itself. Because the persistence proxy requires at least two observed MAP values below 65 mmHg within a three hour window, hypotension events can only be labeled in hours where MAP was actually charted. Patients who are monitored less frequently are therefore less likely to have their hypotension episodes captured by the label at all. The 32 MAP-missing events in the test set are cases where MAP was absent at the time of prediction but was observed often enough in nearby hours for the label to be constructed. Truly unmonitored patients, those with the longest MAP gaps, may experience hypotension that never appears in the dataset. This means the MAP-missing evaluation likely tests the easier end of the MAP-missing population, and real-world performance on the least-monitored patients remains unknown.

## 5.8 Future Work

The most immediate need is external validation. Running the bp_only model with Config G thresholds on an independent ICU dataset (such as eICU [45] or a prospectively collected cohort) would test whether the MAP-missing threshold collapse generalises beyond MIMIC-IV and whether the dual-threshold fix transfers across institutions with different charting practices. If MAP missingness rates differ substantially (for example, 30 percent versus 53 percent), the calibrated thresholds would need to be re-estimated, but the underlying approach should still apply.

Also, subgroup analysis across ICU types, sepsis status, age groups, and monitoring intensity levels would test whether Config G's performance is uniform or varies across patient populations. If certain subgroups have systematically higher or lower MAP missingness, the dual-threshold approach may need population-specific calibration.

Finally, a prospective alarm simulation study, running the system in shadow mode alongside real clinical workflows without influencing care, would provide the strongest evidence for clinical utility. Such a study could measure alarm acceptance rates, clinician response times, and whether earlier warnings lead to earlier interventions, none of which can be assessed from retrospective data alone.

Another direction is to explore adaptive threshold recalibration. In this study, the MAP-missing threshold was calibrated once on the validation set and frozen. In a live system, MAP missingness rates and score distributions may shift over time as monitoring practices change or as the model is deployed in a different unit. An online recalibration mechanism that periodically re-estimates the subset thresholds on recent data could maintain the dual-threshold fix without manual intervention.

It would also be worth investigating whether this coverage disparity problem extends to other EHR prediction tasks where the target variable also serves as the dominant feature. Sepsis prediction models that rely heavily on lactate, for example, may exhibit a similar score distribution collapse during hours when lactate has not been measured. If the pattern generalises, subset-calibrated thresholding could become a standard evaluation step for any alarm system built on intermittently observed clinical signals.

# 6 Conclusion

## 6.1 Thesis Summary

This thesis developed and evaluated an hourly early warning system for sustained hypotension in ICU patients using non-waveform EHR data from MIMIC-IV v3.1. The dataset comprised 94,259 ICU stays on an hourly grid covering the first 48 hours of each admission, with 353 engineered features drawn from 20 clinical signal blocks including blood pressure, vital signs, laboratory results, vasopressor and sedation infusions, respiratory support, fluid balance, and observation-pattern indicators.

Sustained hypotension was defined using a window-based persistence proxy (which defines sustained hypotension as at least two hours with MAP below 65 mmHg within a rolling three hour window), adapted from minute-level benchmarks to tolerate the sparse charting patterns of hourly non-waveform data [15]. All post-onset hours were excluded from training and evaluation to prevent data leakage, and all features were constructed using only data from the current hour or earlier.

Models were evaluated under a fixed 10% alarm budget, thresholds were calibrated on the validation set and frozen before test evaluation. Performance was reported at the hourly level (AUROC, AUPRC, recall, precision), at the event level (event recall, event miss rate), and with operational metrics (lead time, cooldown-adjusted alert burden). Critically, all metrics were stratified by whether MAP was observed at the time of prediction, using three calibration regimes (global, MAP-missing, MAP-observed). Two ablation suites totaling 30 model variants were run across two backends (XGBoost and HGB) to test which clinical signals contribute predictive value and whether non-BP features improve robustness when blood pressure data is unavailable.

A deliberate design choice in the study was that the entire system, from feature engineering to deployment recommendation, is built around lightweight tree-based models (16 to 37 features) that require no GPU and no specialised infrastructure, with very minimal hardware requirements. This matters because the clinical settings where MAP is most likely to be absent (facilities without continuous arterial line monitoring, step-down units, hospitals relying on intermittent cuff readings) are also the settings least likely to support deep learning inference pipelines. The model's simplicity is matched to the deployment context where the MAP-missing problem is most acute.

## 6.2 Key Findings

Seven main findings emerged from this work.

Blood pressure features dominate hypotension prediction at the three-hour horizon. The bp_only model achieved an AUROC of 0.955 and an event miss rate of 0.66% in the compressed suite using 37 features. BP-containing variants with small additions, such as bp_plus_gcs and bp_plus_resp_support, performed similarly, but no richer feature combination provided a clear practical improvement over bp_only. Larger expansions, such as bp_plus_all_new, scored lower than bp_only across the main reported metrics.

Non-BP features plateau at the information ceiling of approximately 0.75 AUROC. Six progressively richer composites (from 161 to 319 features) all scored between 0.746 and 0.748. Within this study, non-BP features plateaued regardless of model capacity or feature count, suggesting a signal limitation for this particular dataset, label definition, and prediction horizon, though this conclusion cannot be generalised without testing alternative task formulations.

A globally calibrated alarm threshold produces zero alarms for 55.7% of patient hours. When MAP is absent, the bp_only model's scores collapse to near zero (mean = 0.012), and the global threshold (0.586) sits above 99.994% of those scores. All 32 MAP-missing events in the test set received no warning. Standard aggregate metrics conceal this failure entirely.

Subset-calibrated thresholds on the same model recover 93.75% recall on MAP-missing hours (95% CI: 79.2% to 99.2%, n = 32 events). The wide interval reflects the small number of MAP-missing events in a single-institution test set; the direction of the result, from zero coverage to strong coverage, is robust even at the lower bound. The model's ranking ability remained intact as measurement timing features (hours_since_MAP ranked third in permutation importance) carry predictive signal even when MAP values are absent. The fix requires no retraining and no additional model.

The proposed configuration (Config G) uses one model with dual thresholds and is the best-performing configuration in this retrospective evaluation. It achieves 97.6% overall event recall and 93.8% MAP-missing event recall at 0.89 alerts per 24 hours under a six hour cooldown. The model outperforms all gated two model alternatives, which max out at 25% MAP-missing event recall at matched alarm rate. Config G also requires no model switching logic, no ensemble infrastructure, and no GPU. It is a single tree model with a threshold lookup conditioned on MAP availability. Its computational simplicity means it could, in principle, run in the clinical settings where MAP absence is most common, such as units without continuous arterial monitoring, though multi-setting prospective validation would be needed before any deployment recommendation could be made

Event-level evaluation reveals that models perform better than what hour-level metrics would suggest. The model catches 82.2% of positive hours but 98.3% of events, because events span multiple hours and a single alarm suffices for detection. Event-level metrics more directly reflect clinical utility and should be reported in ICU early warning studies.

## 6.3 Recommendations and Takeaways

Several recommendations and takeaways from this work, directed at work realizing the construction of an early warning system for hemodynamic instability using intermittently charted EHR data.

MAP-missing hours need explicit handling. Any alarm system that depends on blood pressure features will produce systematically lower risk scores when blood pressure has not been measured recently. A single globally calibrated threshold silently overlooks these hours, regardless of how strong the model's aggregate performance appears. The simplest fix is Config G, calibrating one threshold for MAP-observed and a second, lower threshold for MAP-missing hours, both targeting the same alarm rate within their respective subsets. This needs no second model as it is a model threshold lookup table modification.

Another is to report event-level metrics alongside hourly metrics. Hour-level recall alone understates clinical utility because hypotension events span multiple hours and a single alarm before onset is sufficient. The event miss rate at a fixed alarm budget is a more direct measure of whether the system would actually help a clinician. Studies that report only AUROC and hourly sensitivity leave the clinically relevant questions unanswered.

A suggestion is to encode missingness rather than to impute it away. Observation-pattern features (such as binary observation flags, hours-since-last-observation counters, within-window measurement counts) ranked among the most important predictors in both the only blood pressure and non blood pressure models. The measurement timing encodes clinical concern, monitoring intensity, and proximity to the last known patient state [29]. Imputing or forward-filling these patterns erases features for model to use.

Finally, to keep models simple. The proposed model's simplicity also helps in clinical settings. The blood pressure only model outperformed every richer variant tested, including composites with 308 features. The tree-based architecture runs on commodity hardware with no GPU requirement. This matters as the clinical settings where MAP-missing hours are most common, such as hospitals without continuous arterial line monitoring, step-down units, and facilities relying on intermittent cuff readings, are also the settings least likely to have the infrastructure needed for heavier architectures like LSTMs or Transformers. Config G is operationally viable precisely where it is needed most, a single lightweight model with a threshold lookup, running on any machine that can query an EHR.

This thesis demonstrates that effective early warning for sustained hypotension requires more than a well-performing model. It requires careful attention to how the event is defined under sparse data, how the alarm threshold behaves across different monitoring regimes, and whether the evaluation framework tests what actually matters for deployment. A model that achieves 0.955 AUROC can still fail silently for more than half of all patient-hours if these questions are not asked. Asking them, and showing that the answer can be solved via lightweight architectures, is the contribution of this work.

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